

Box Robot Base

version 1.0, rev 15 27 December 2025

This is a classic box robot design intended to provide two forward facing attachments or a combination of forward and sidefacing attachments. We usually configure the left attachment motor for our worm drive attachment and the right attachment motor, rotated at 90 degrees, is usually for geared lifting arm. We also use the right motor to drive a forward facing axles to turn objects on the FLL board (like the gears in Masterpiece Mission 12 rotating the chicken or Unearthed Mission 11/Angler Artifact).

When assembling the robot care needs to be taken on how the sensor and motor cables need to be routed prior to joining the front and rear sections. Then secure the cables using the wire connector/49283 in rear compartment containing the hub.

It is loosely based on:

<https://www.youtube.com/watch?v=PCMEvf4p2PI&t=2s>



116x
2780
Black



2x
37308c01
Black



2x
56908
Black



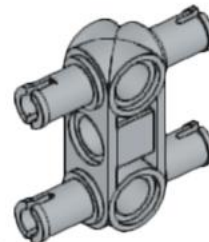
5x
73507
Black



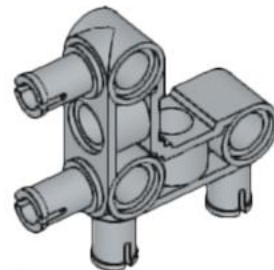
2x
32073
Black



4x
32525
Black

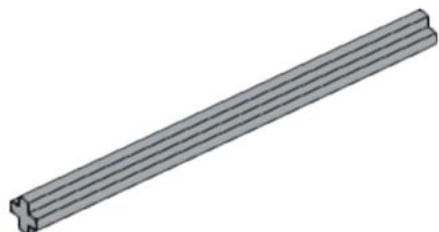


2x
48989
Light Bluish Gray

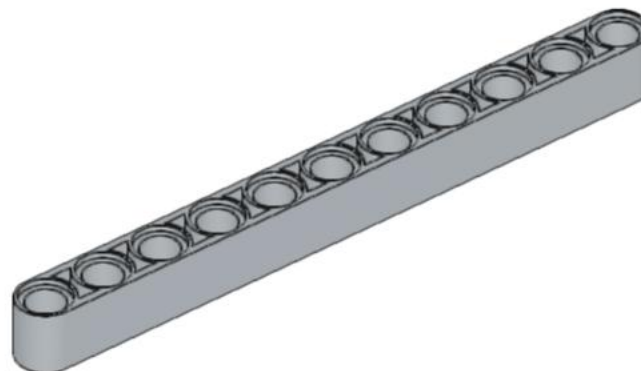


2x
55615
Light Bluish Gray

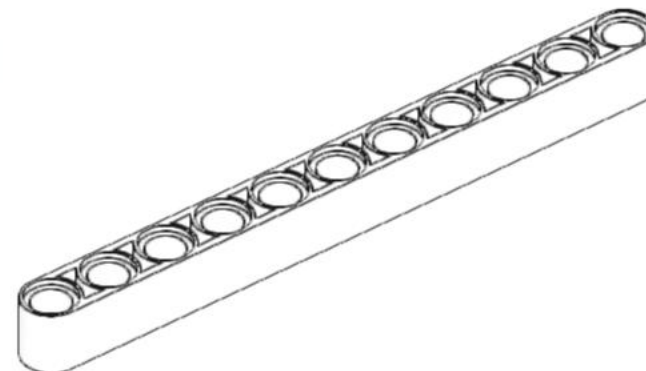
You might want to use a 4L axle instead of the 5L.



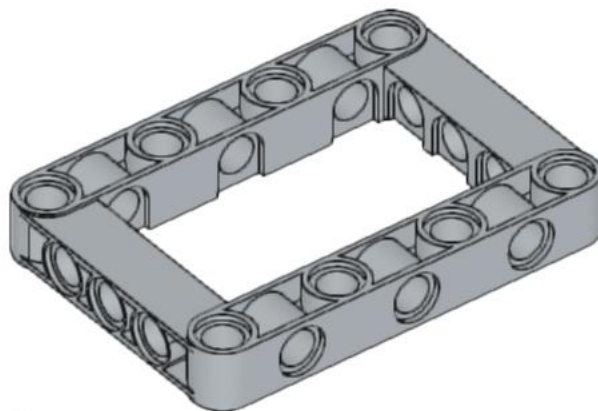
2x
44294
Light Bluish Gray



3x
32525
Light Bluish Gray



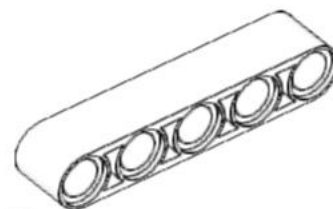
1x
32525
White



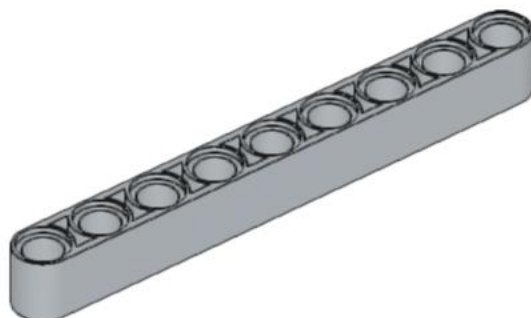
4x
64179
Light Bluish Gray



4x
18654
White



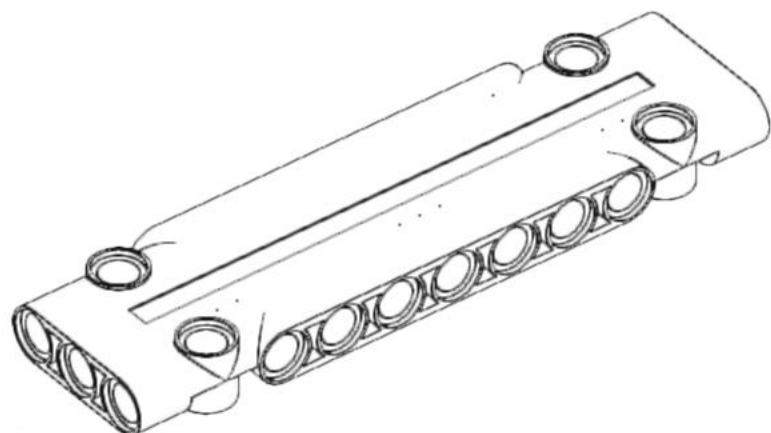
2x
32316
White



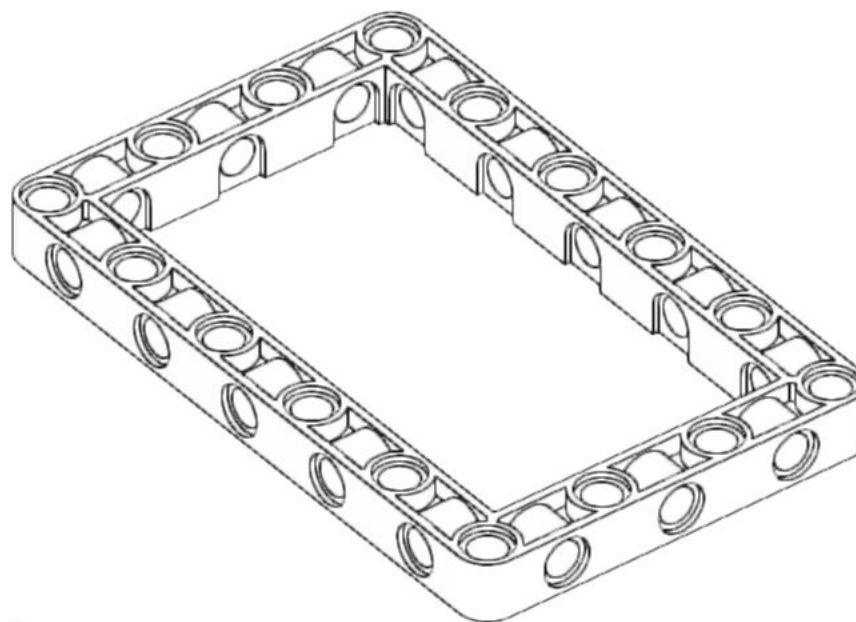
8x
40490
Light Bluish Gray



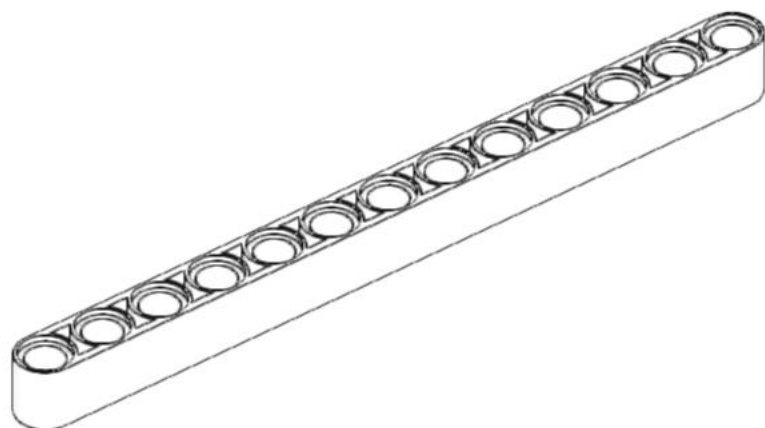
2x
32020c01
White



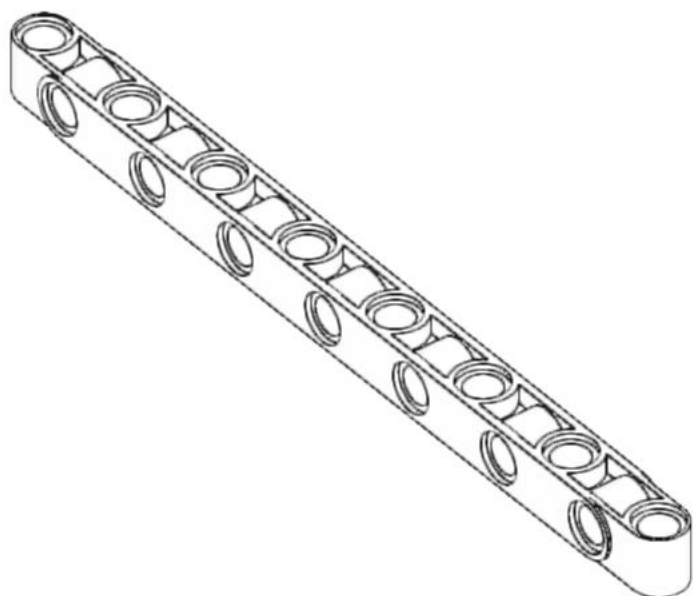
2x
15458
White



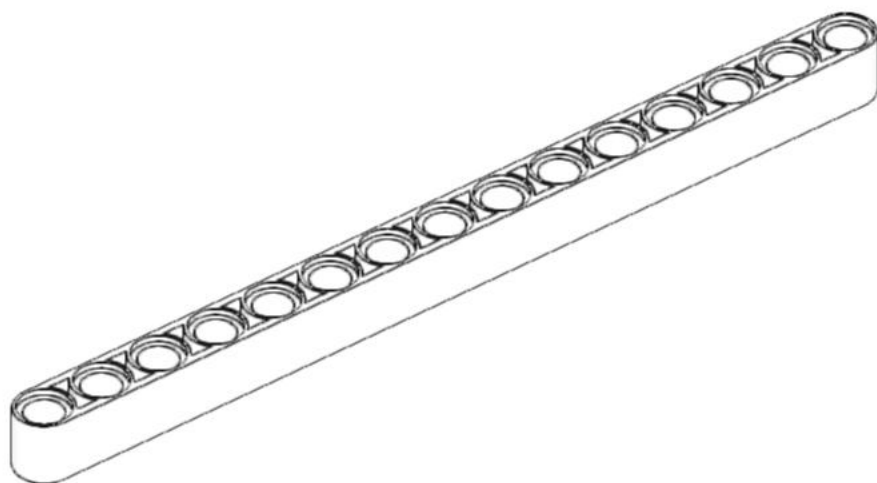
3x
39794
White



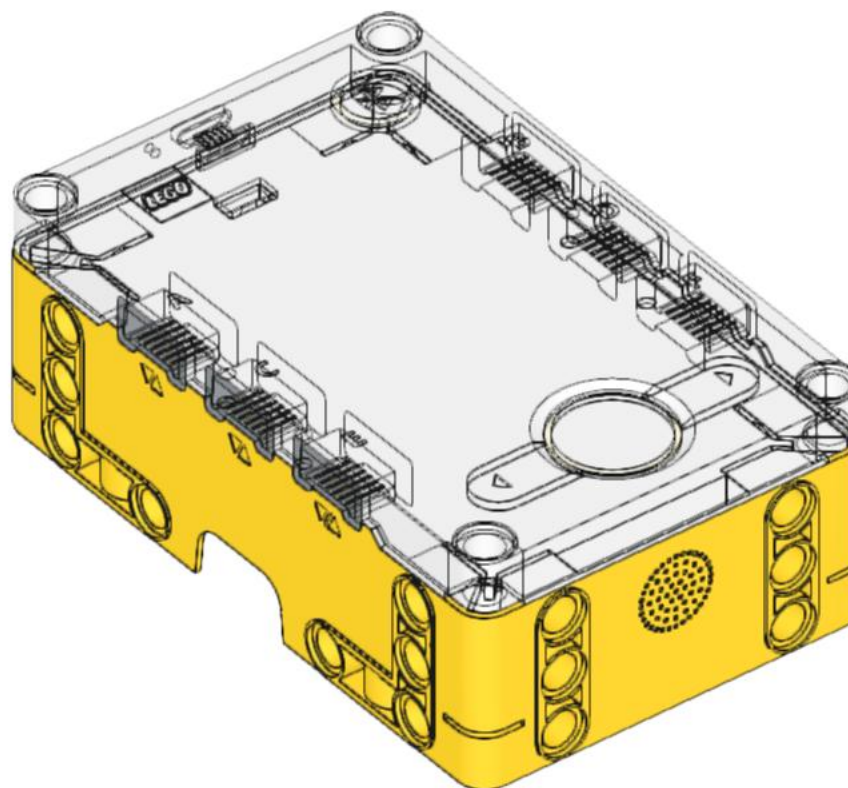
2x
41239
White



1x
71710
White



1x
32278
White



1x
53444c01
Yellow



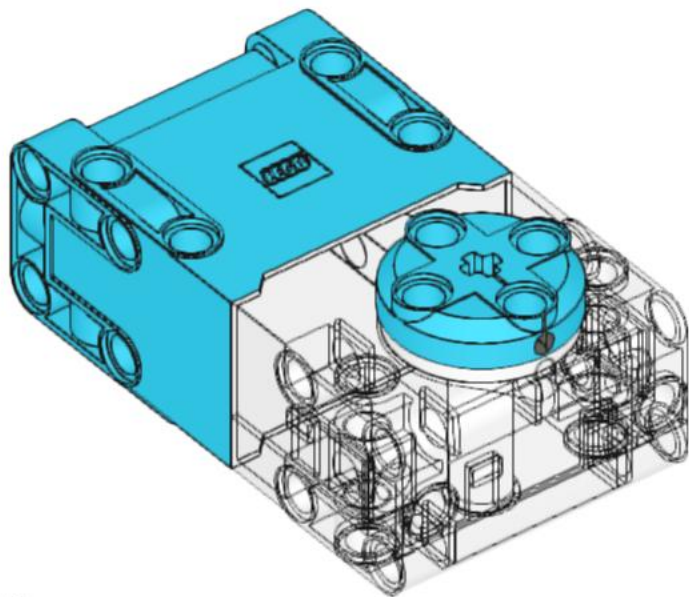
5x
39369
Yellow



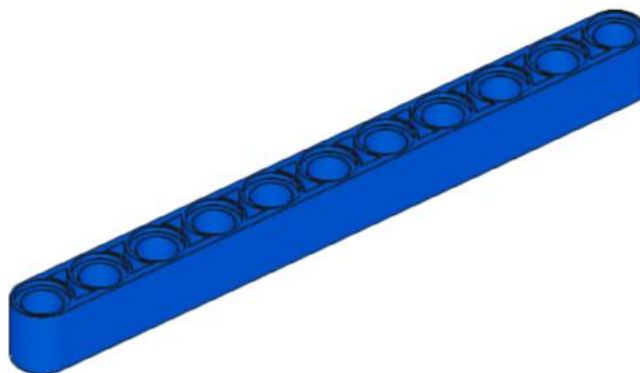
2x
4265c
Blue



81x
6558
Blue



2x
54675c01
Medium Azure



2x
32525
Blue

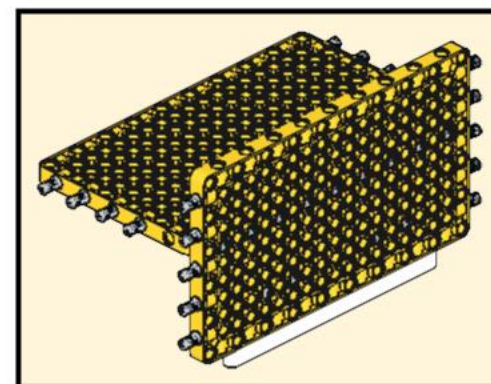
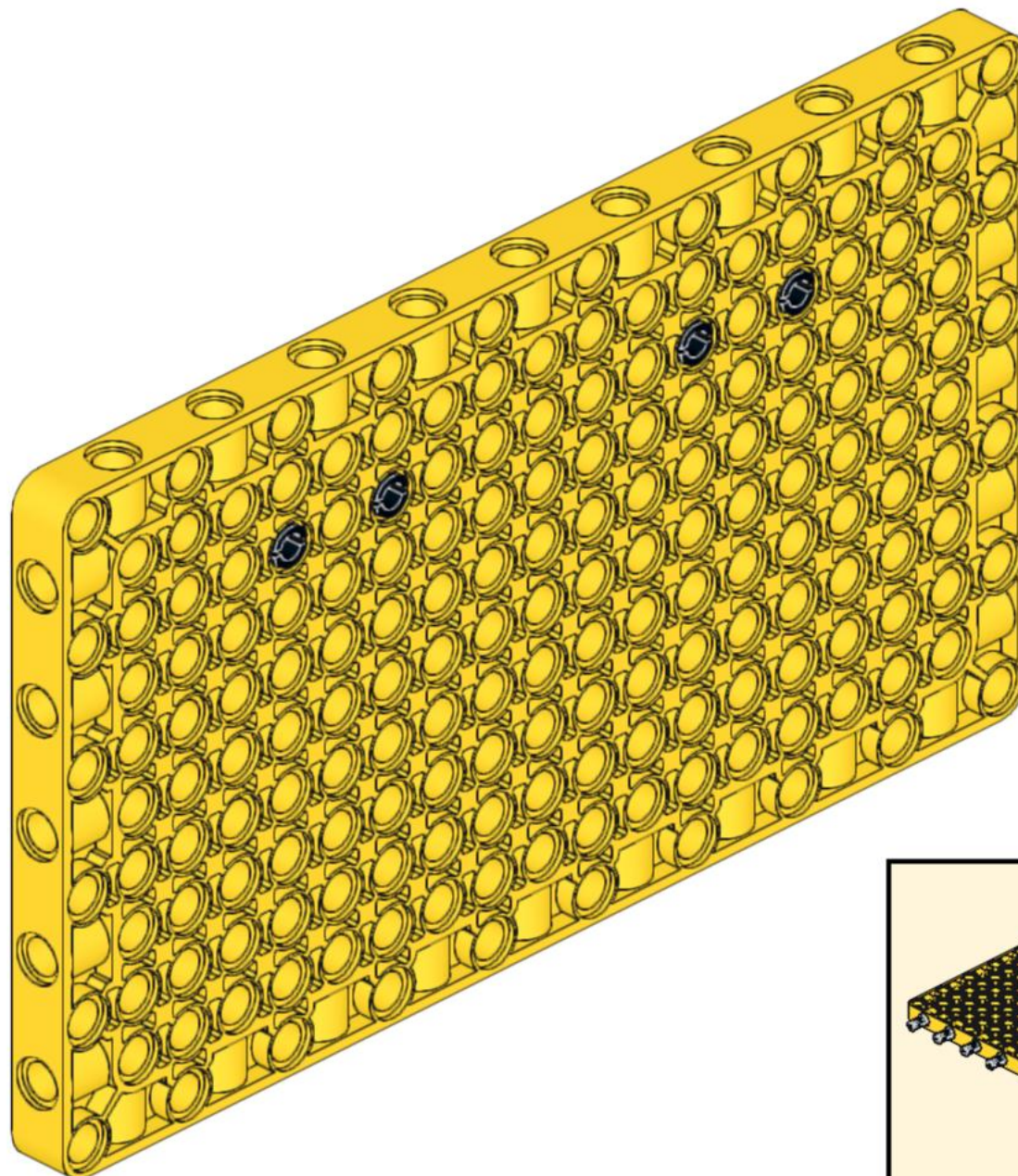
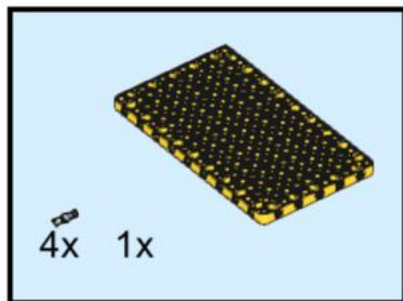
Front Assembly

Be careful on how attachment motors are placed on top of the yellow plate covering the driving motors.

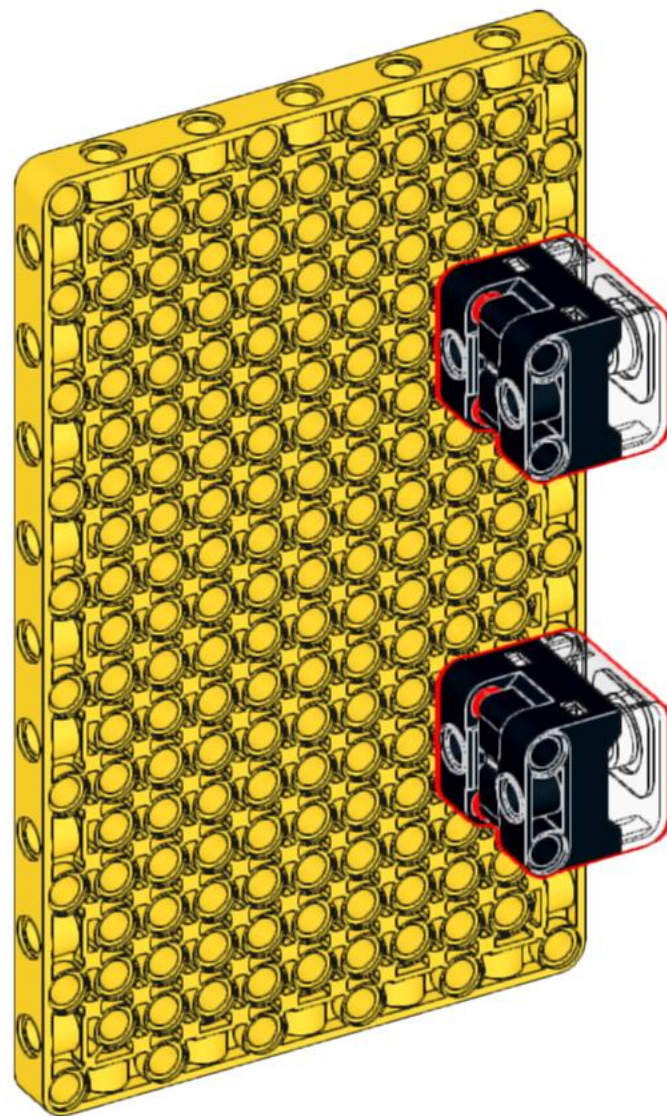
The primary mechanism the drive motors are secured is via attachment to this plate.

You need to optimize the number of pins (8) from the drive and attachment motors that are connected to the top and bottom of the plates.

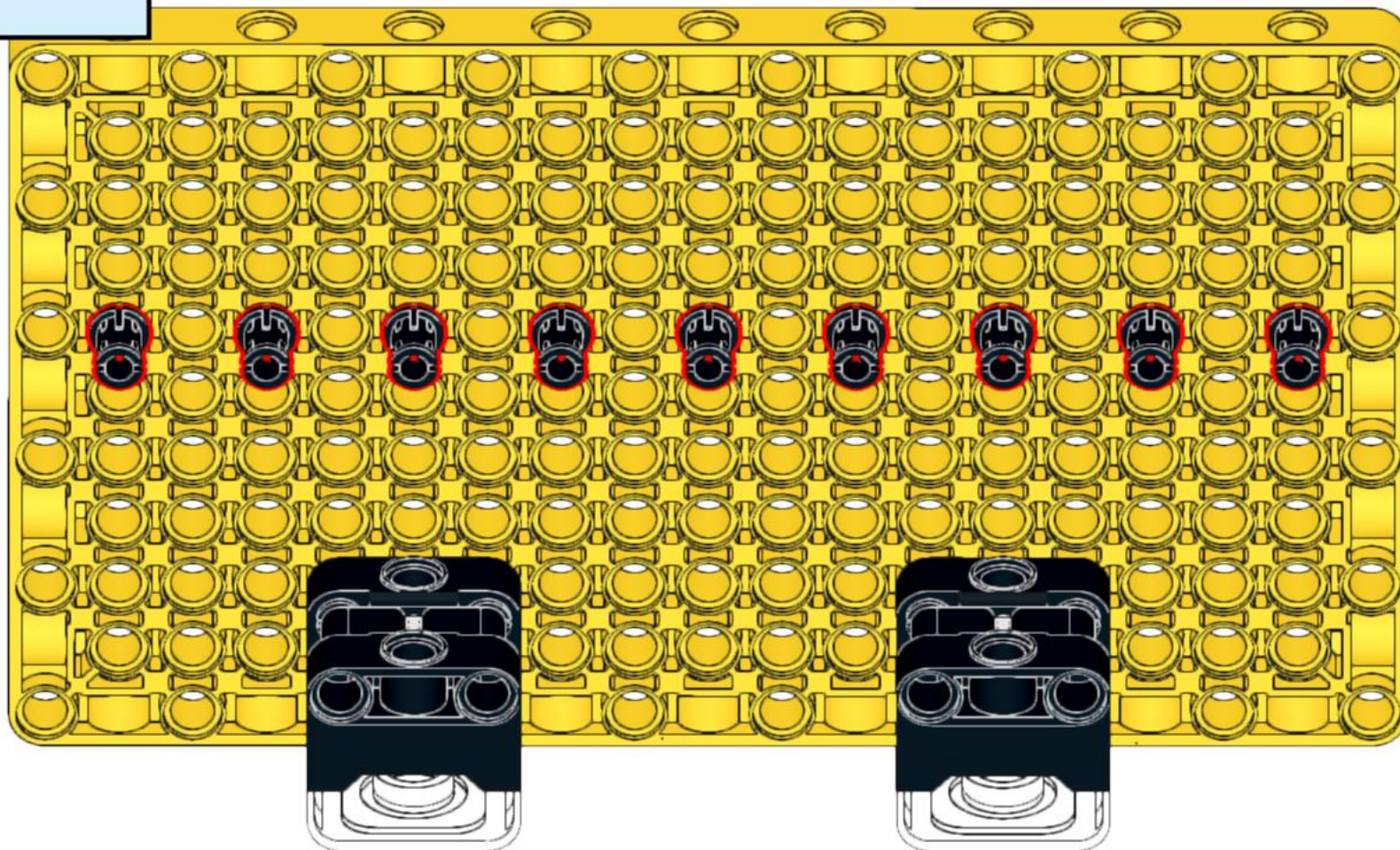
1



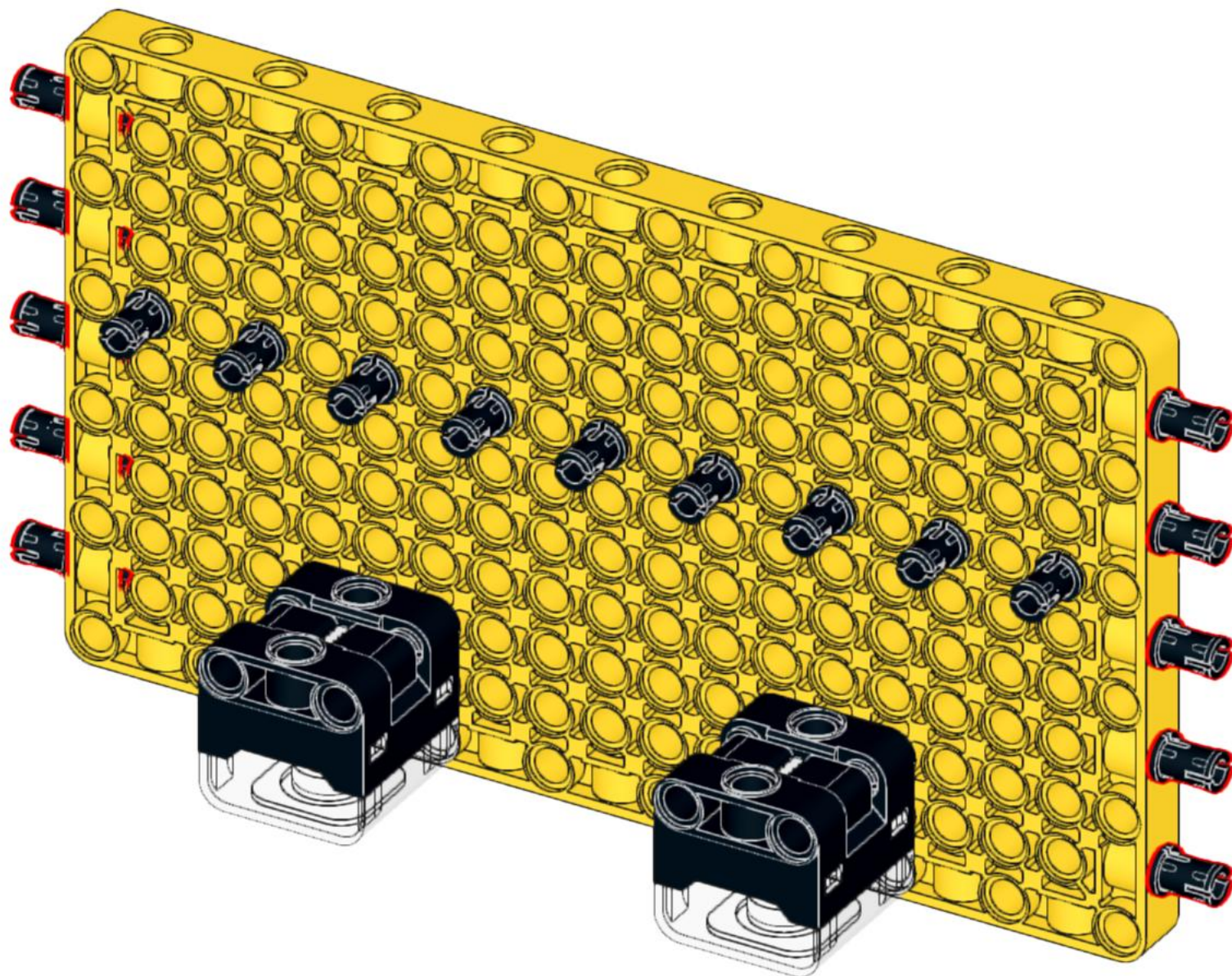
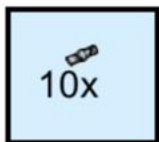
2



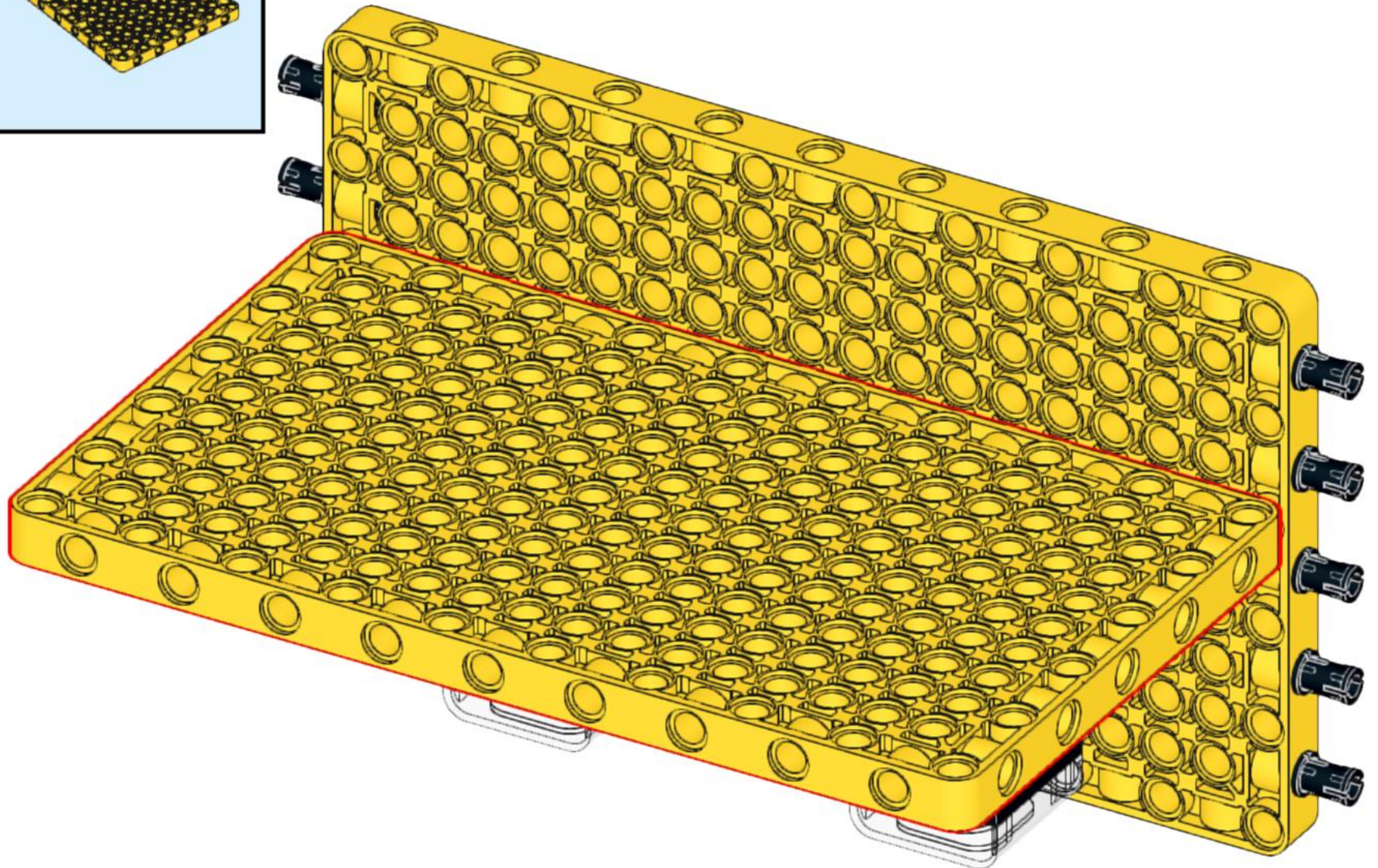
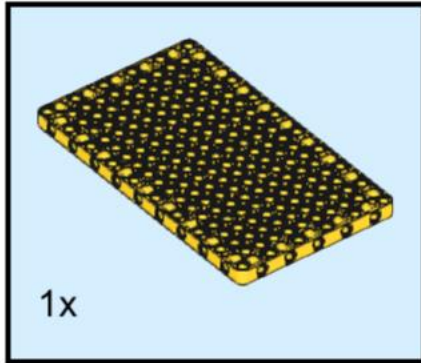
3



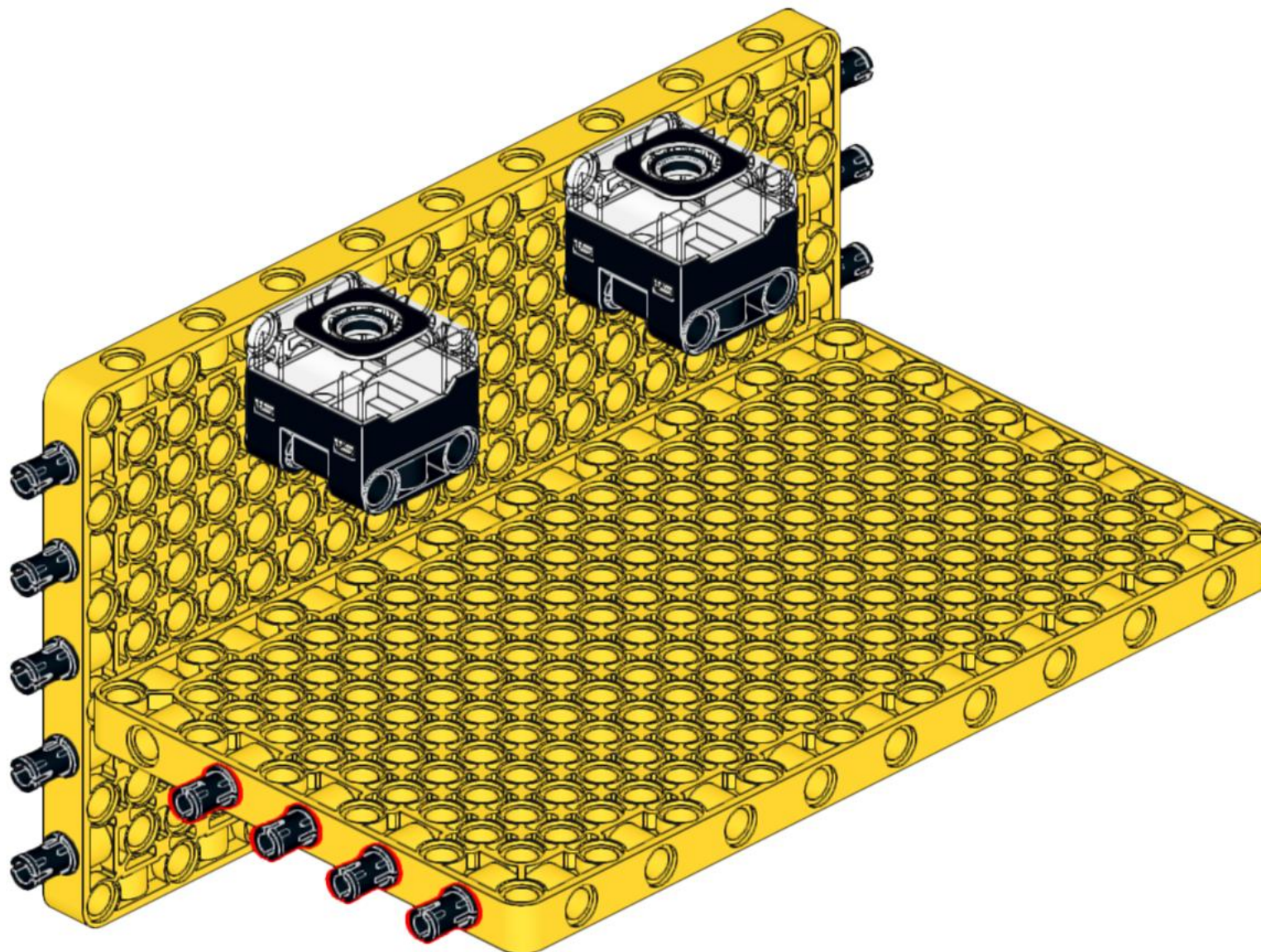
4



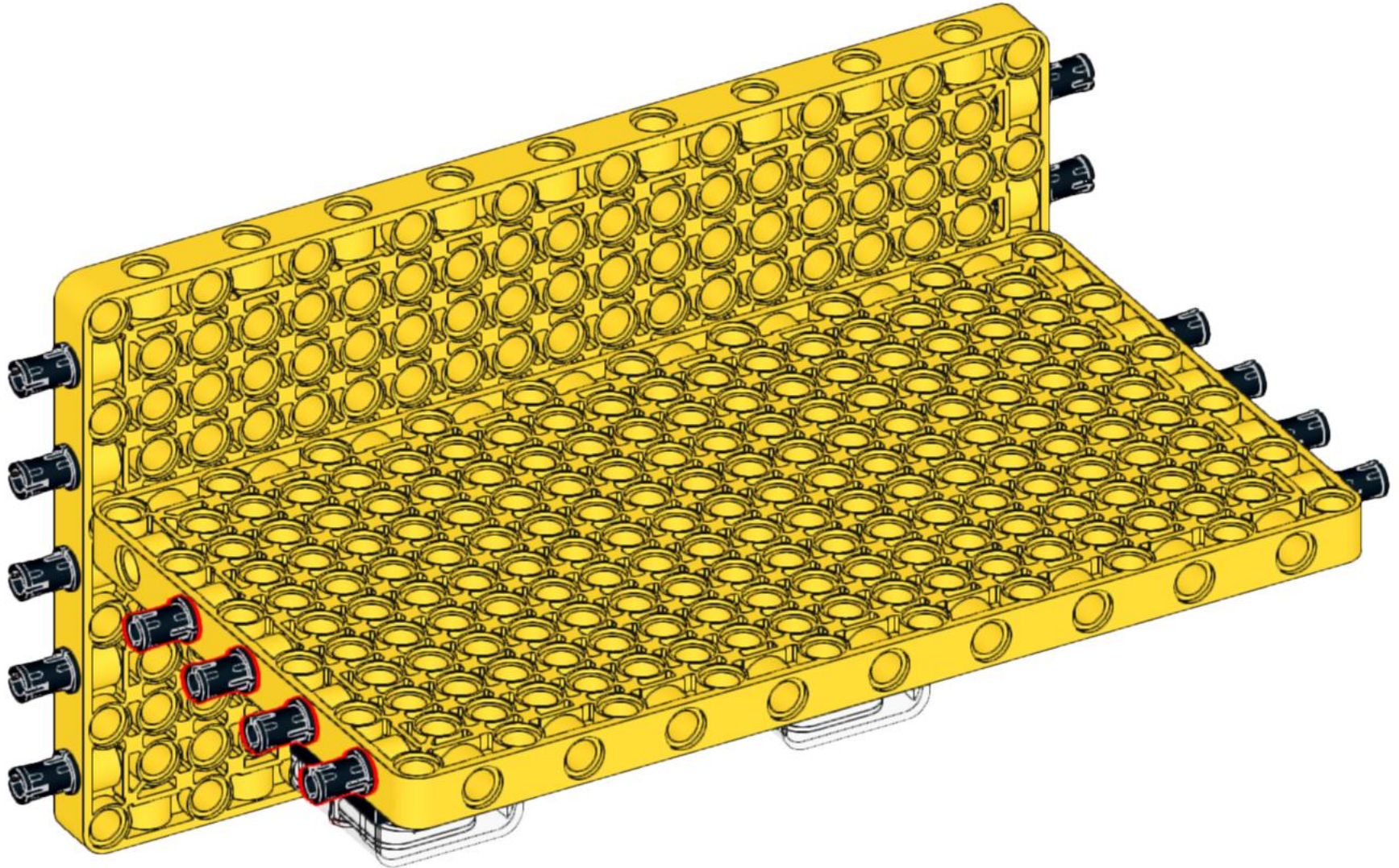
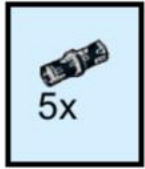
5



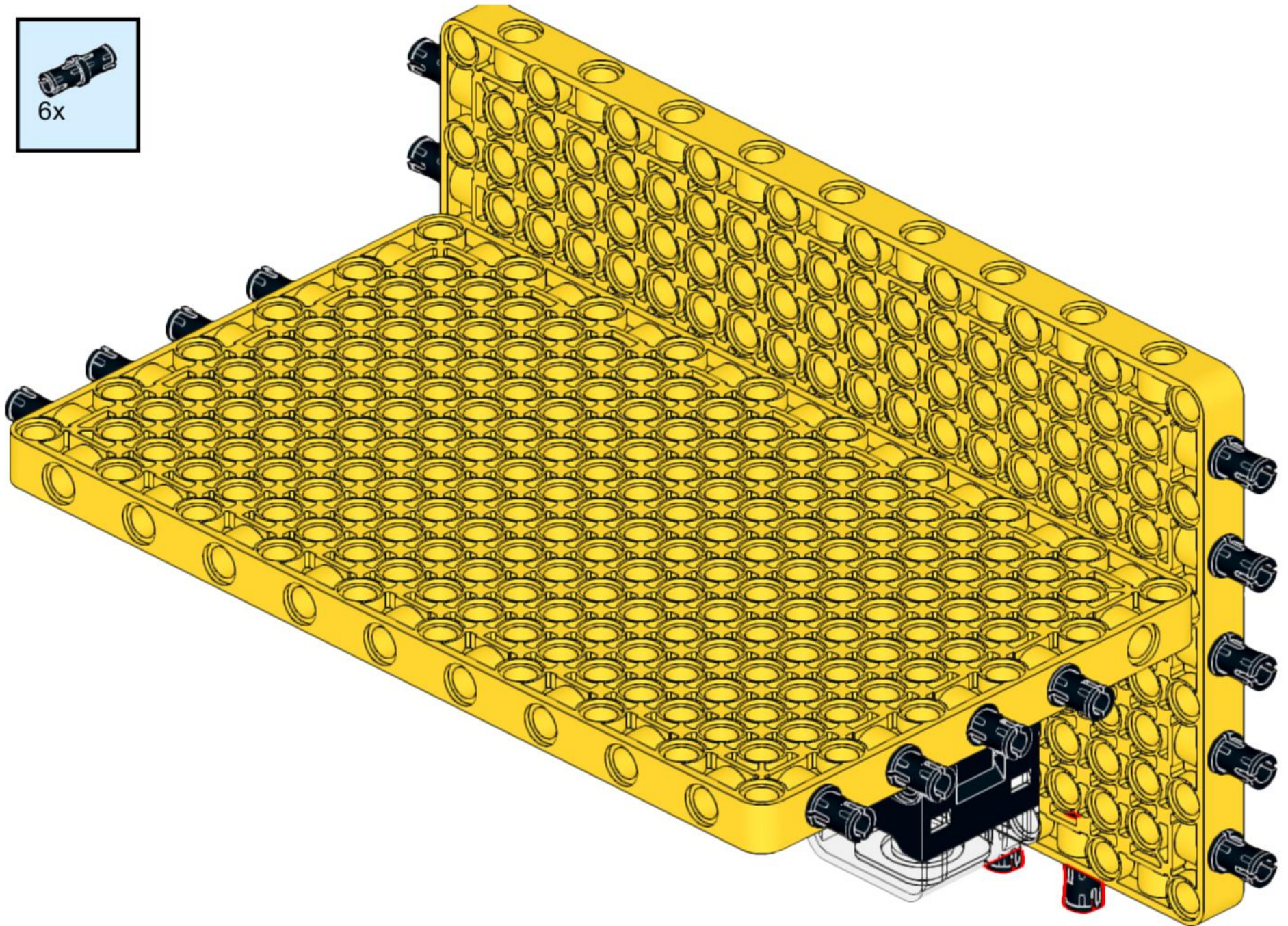
6



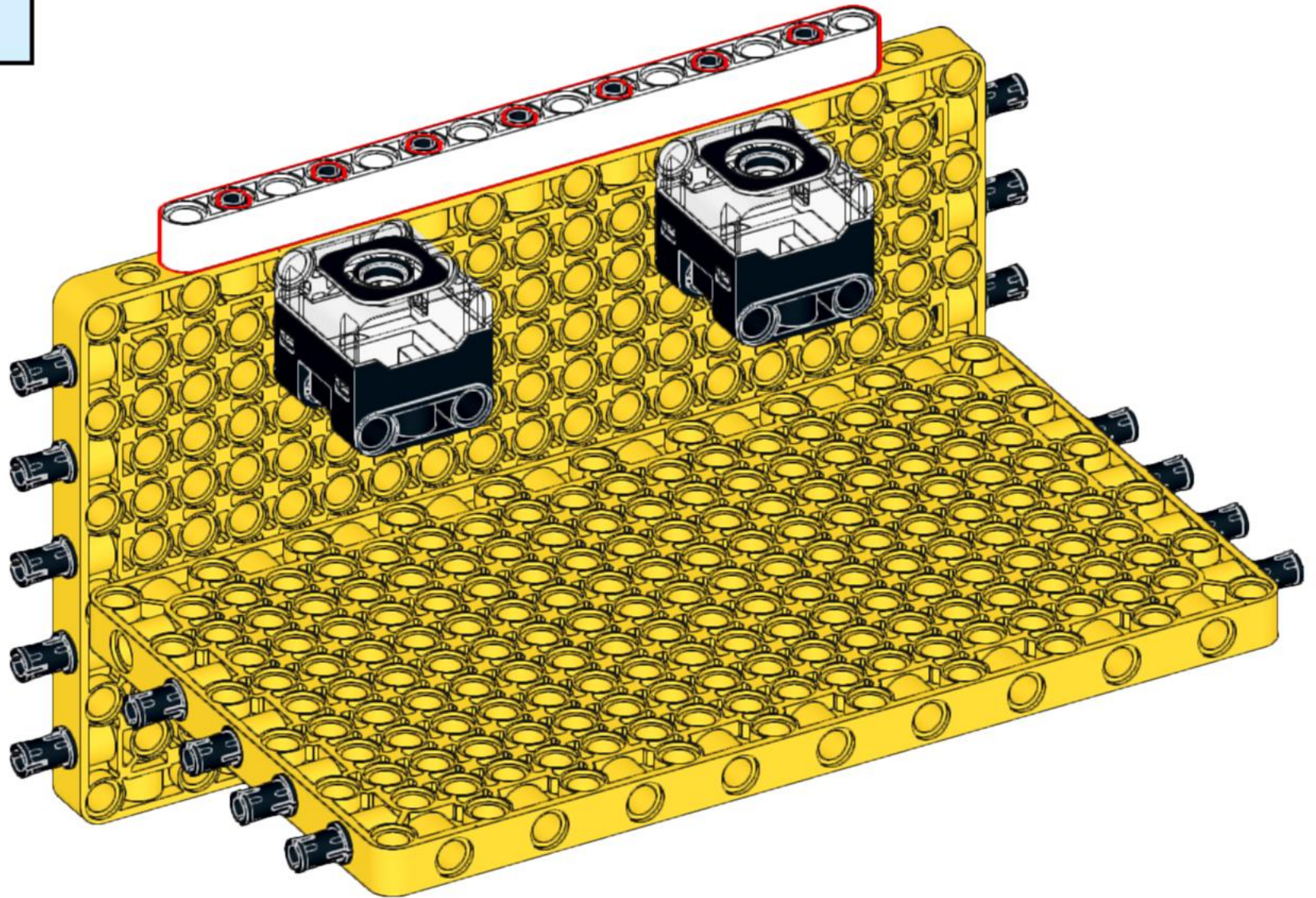
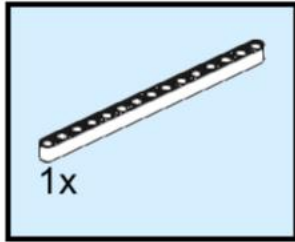
7



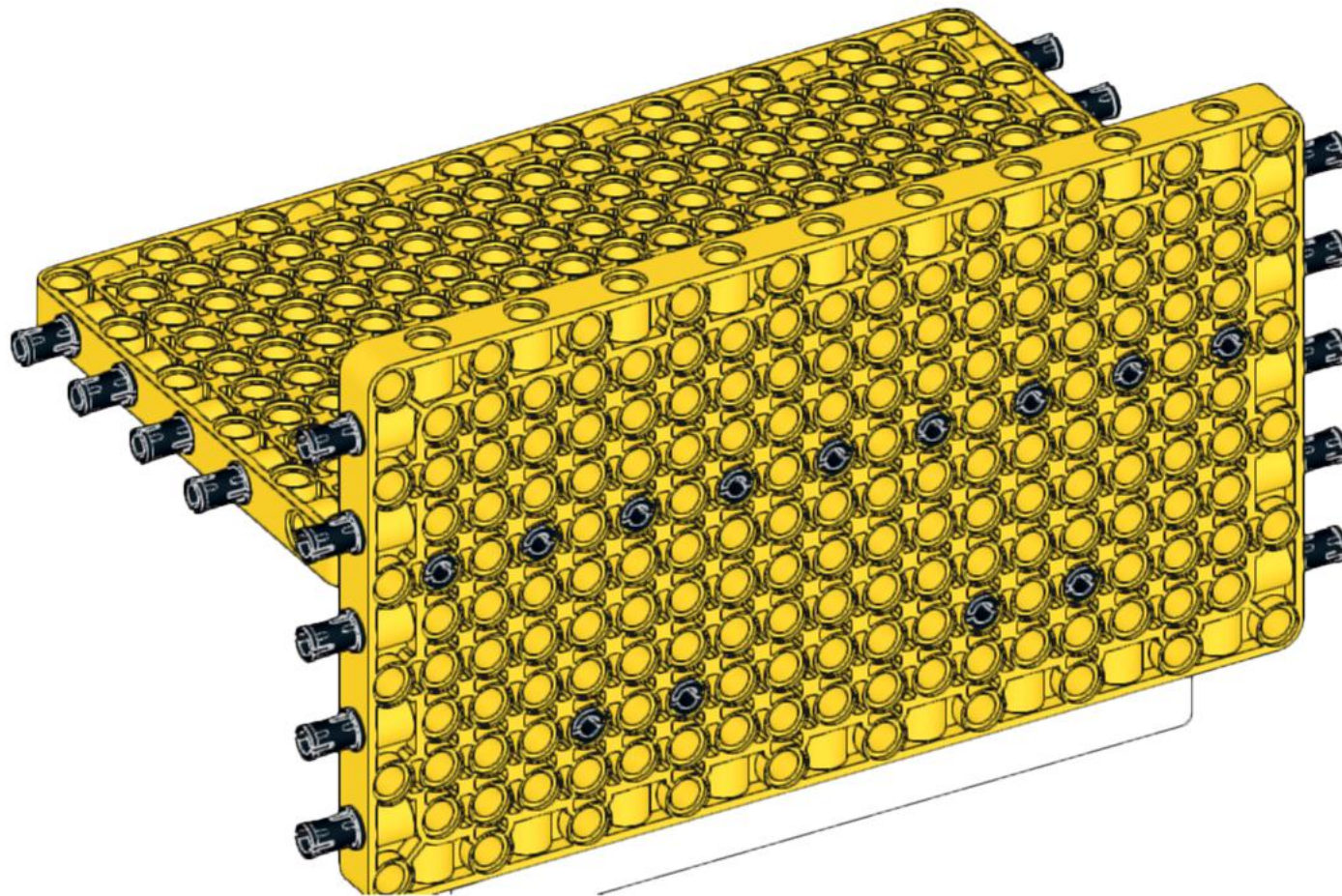
8



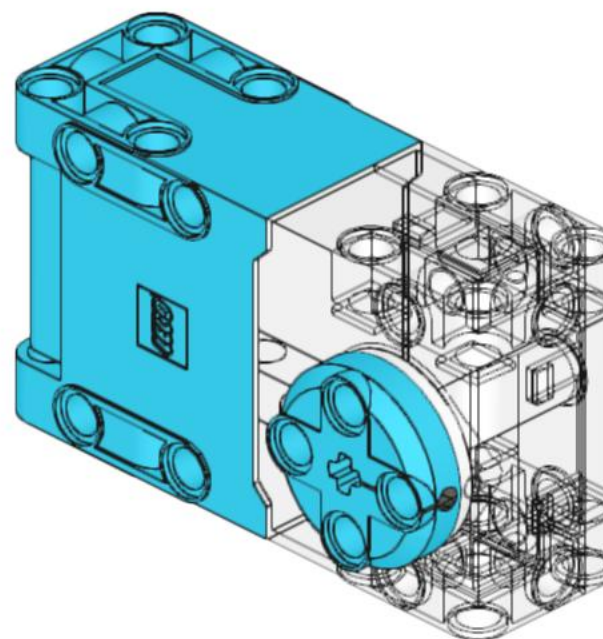
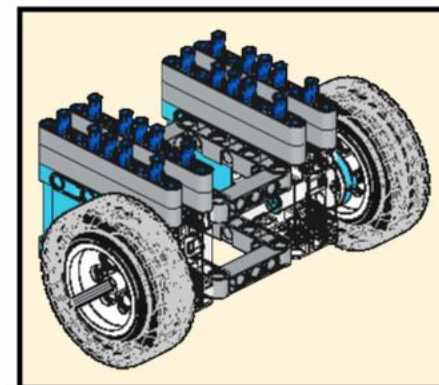
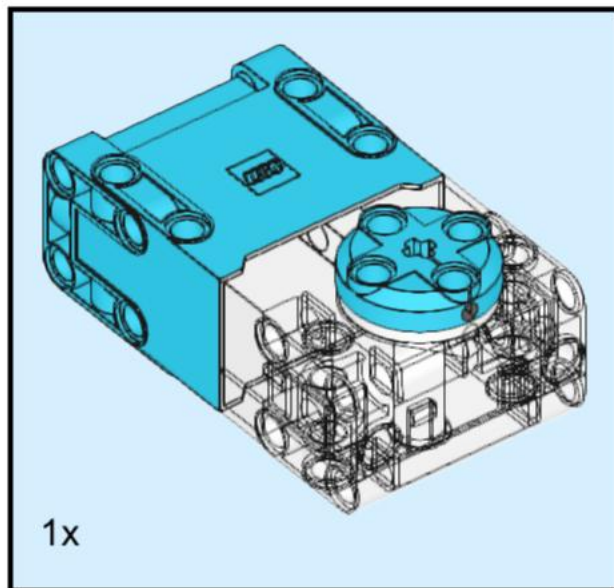
9



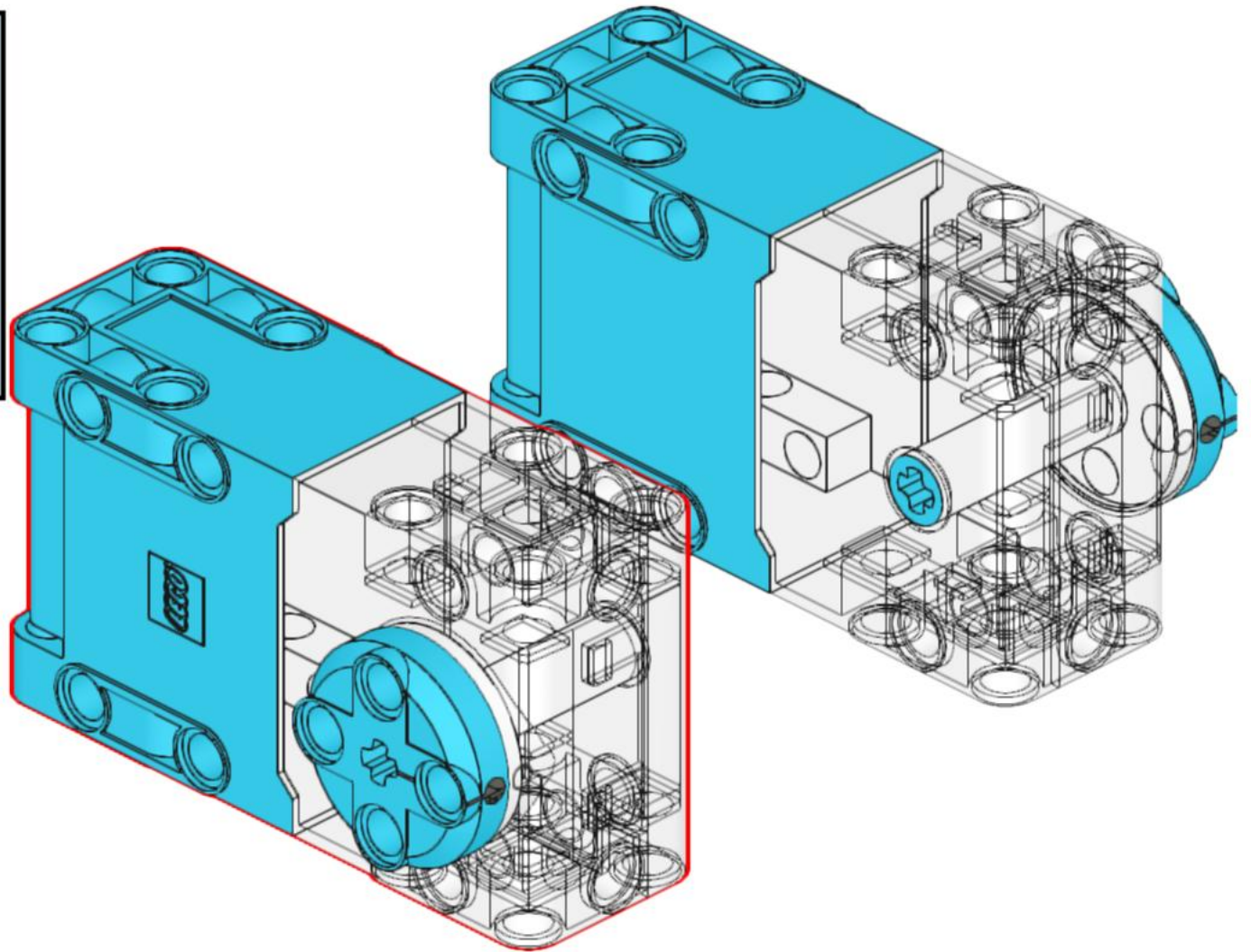
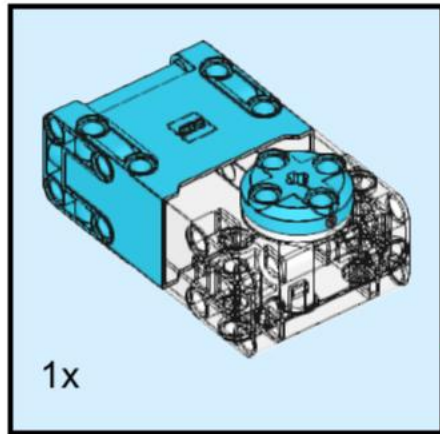
10



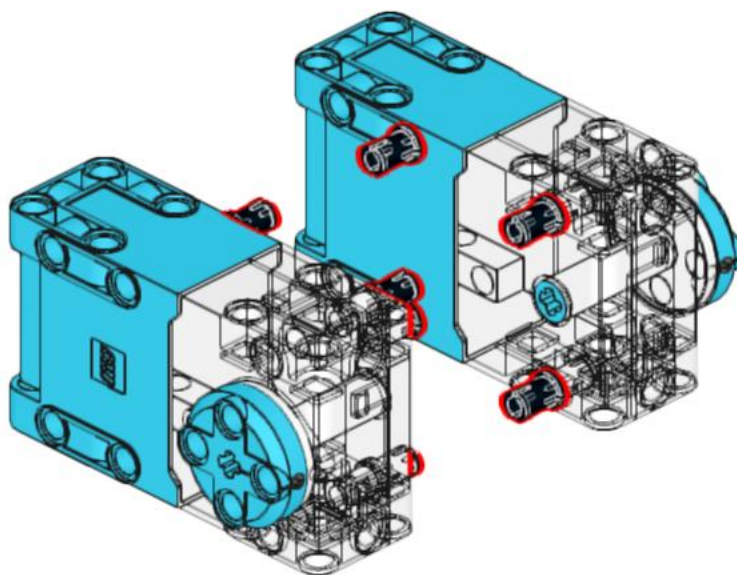
11



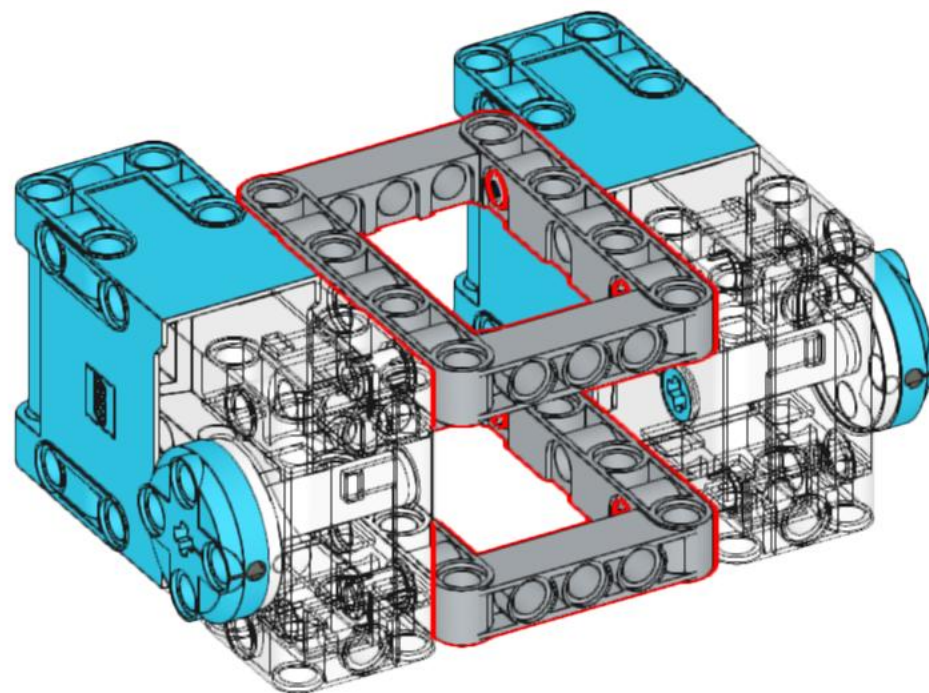
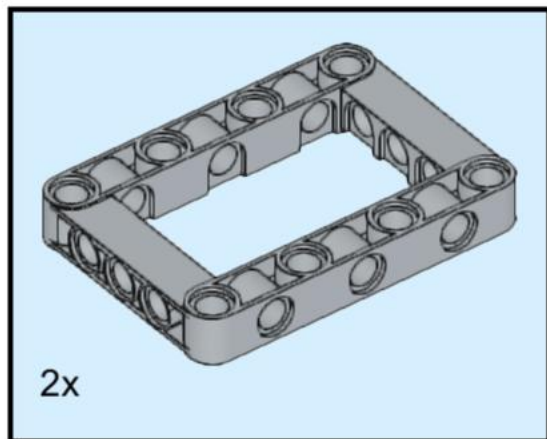
12



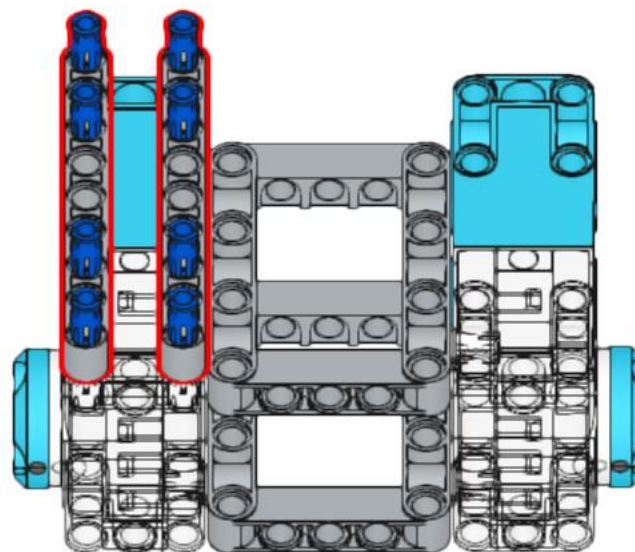
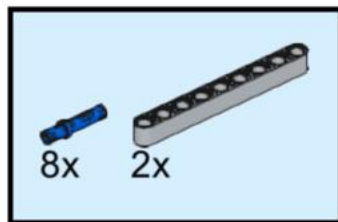
13



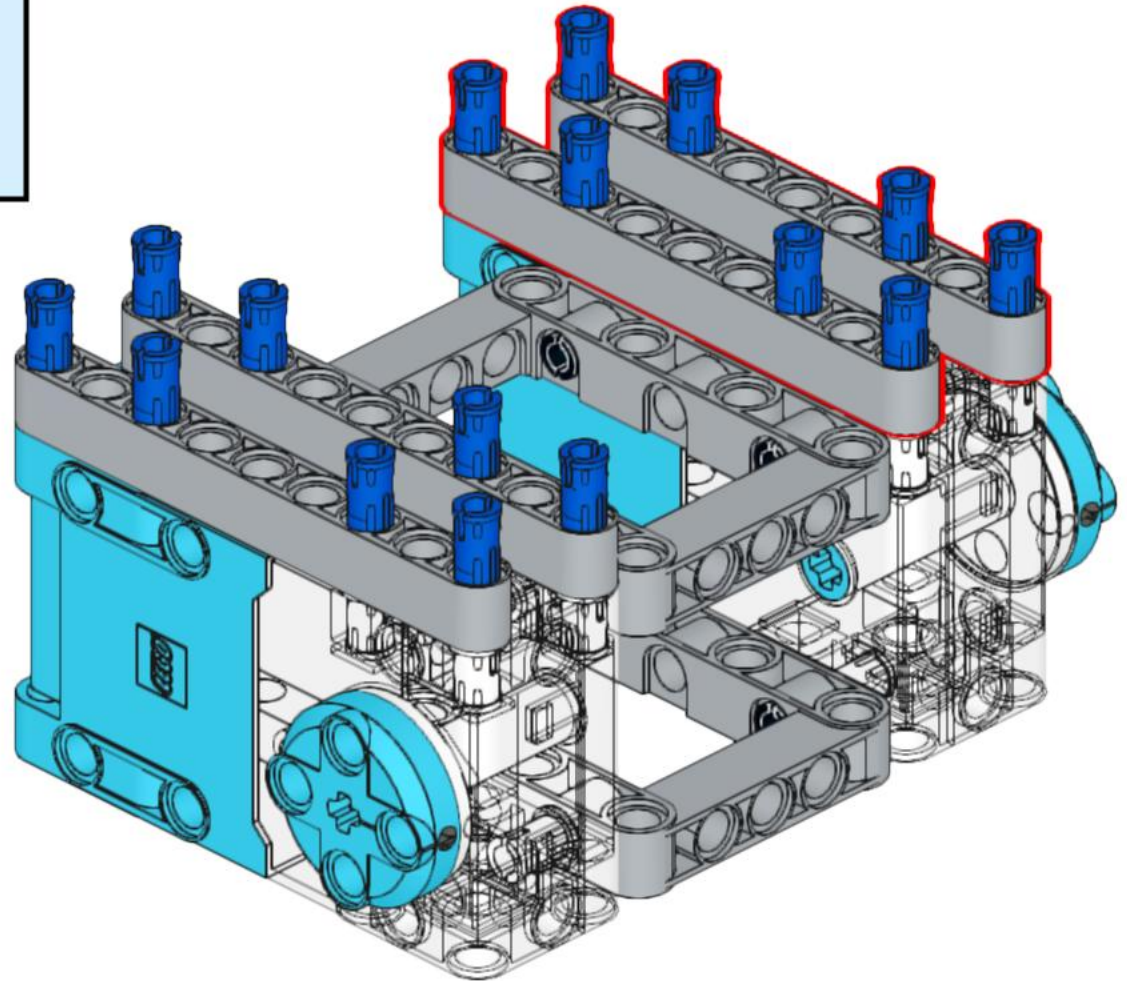
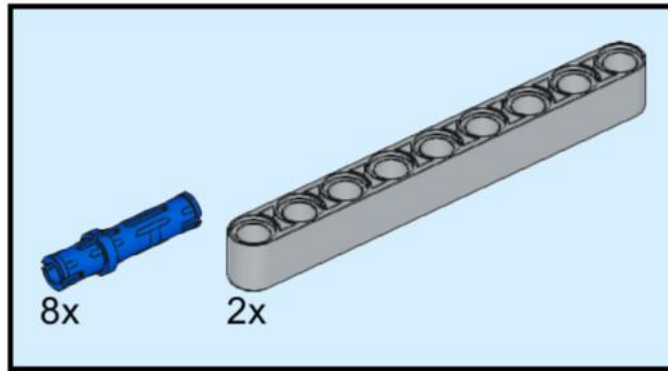
14



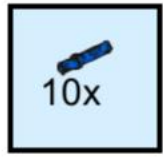
15



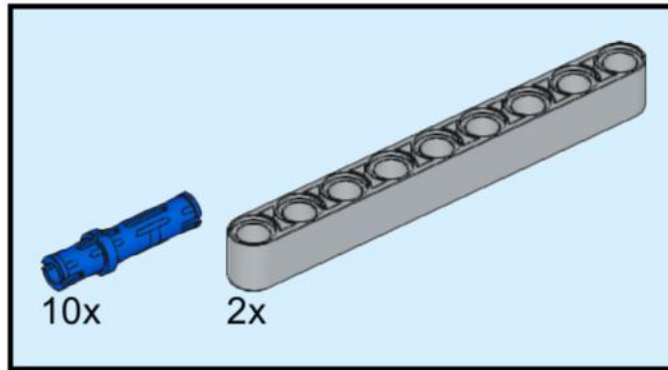
16



17



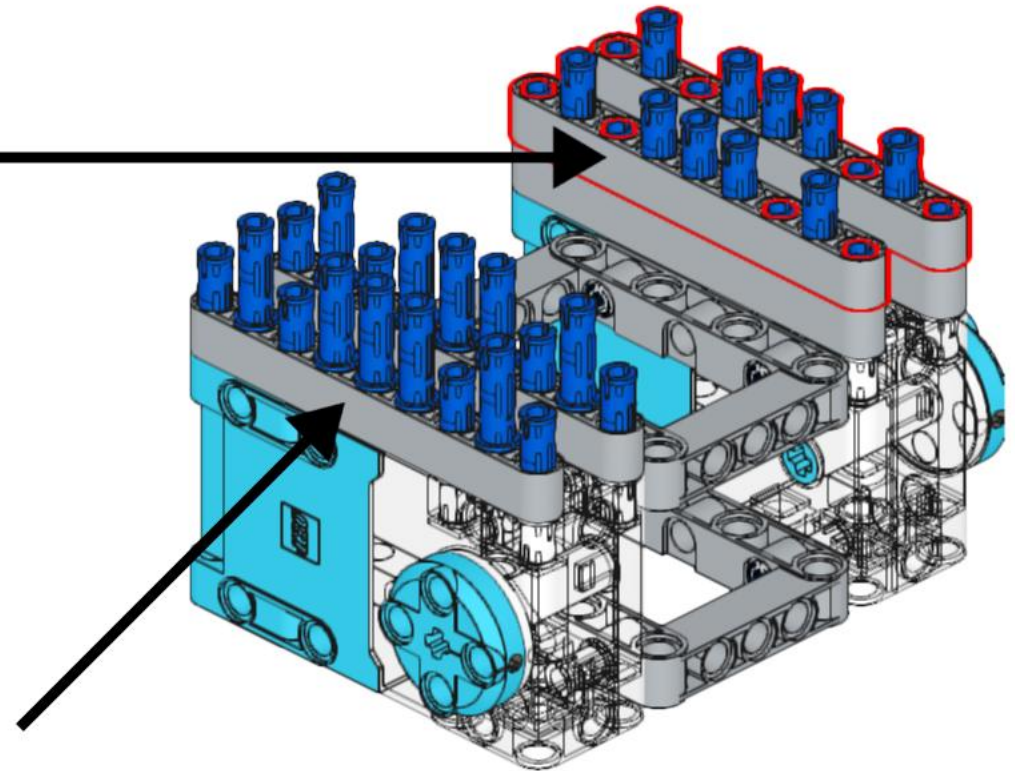
18



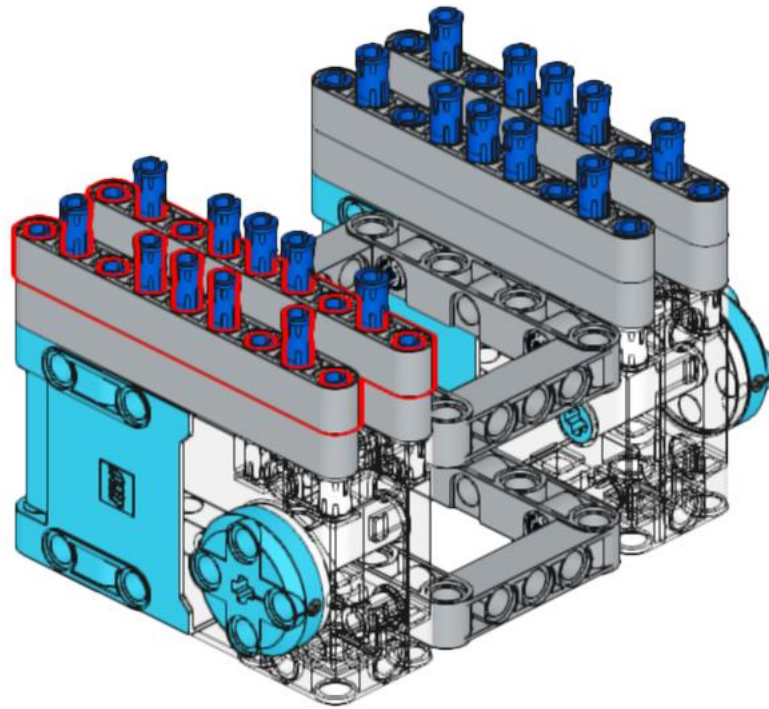
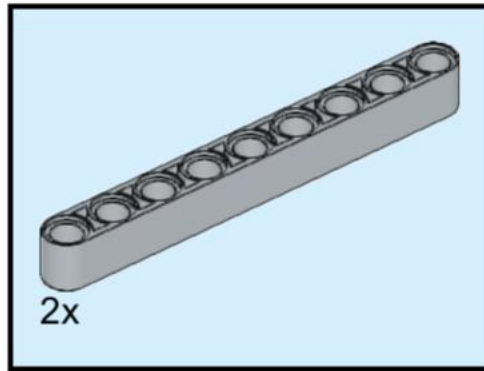
You might need to change which 3L pins are present depending on how your left and right attachments align up on the 11x19 plate that sits on top of the drive motors.

Another option is instead of recessing all the 3L Pins into the bottom gray beam, you can instead connect them to the top gray beam so they will penetrate the 11x19 plate and connect to your attachment motors.

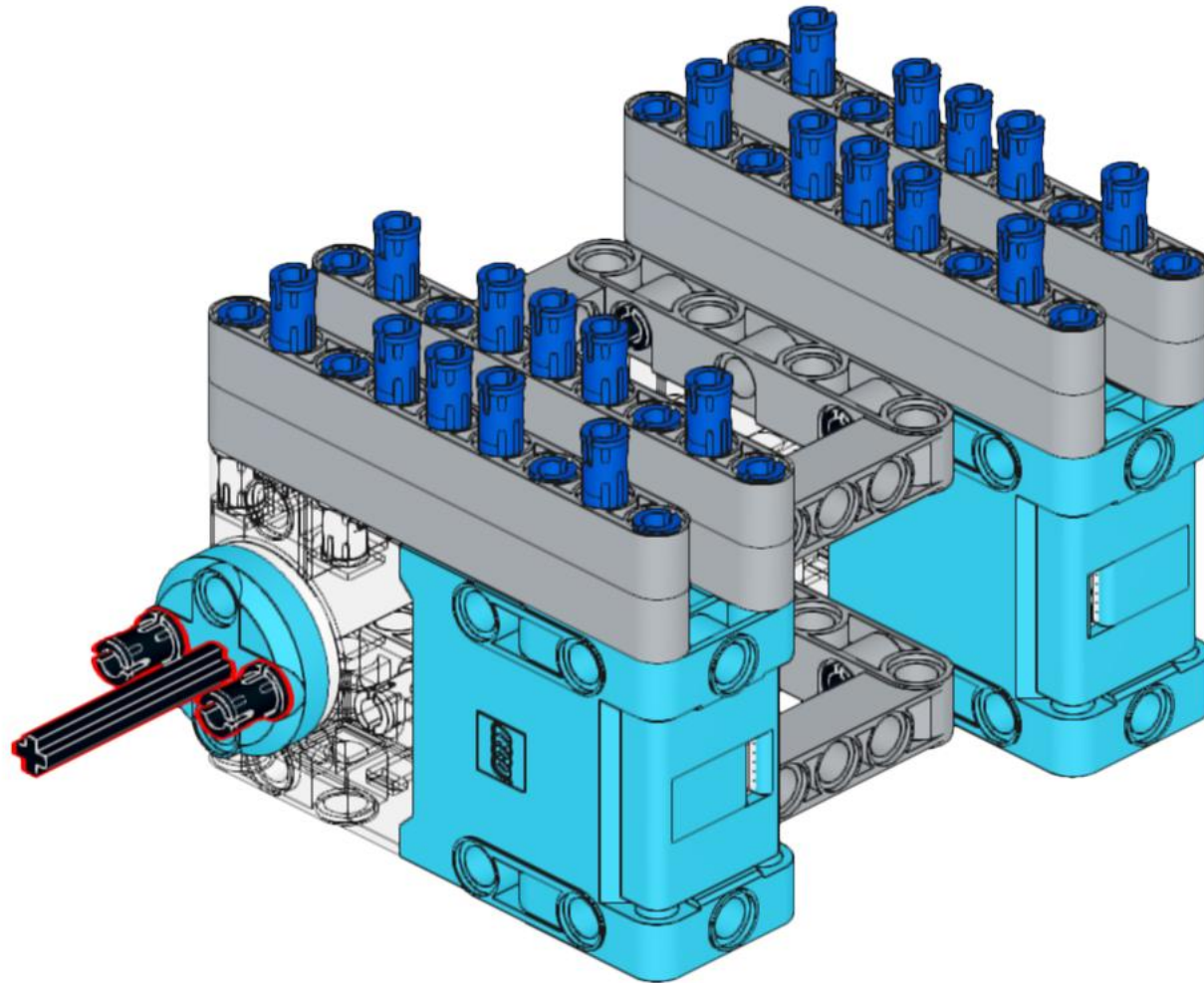
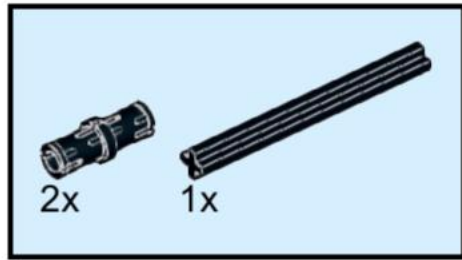
It is a tradeoff in strength vs. flexibility.



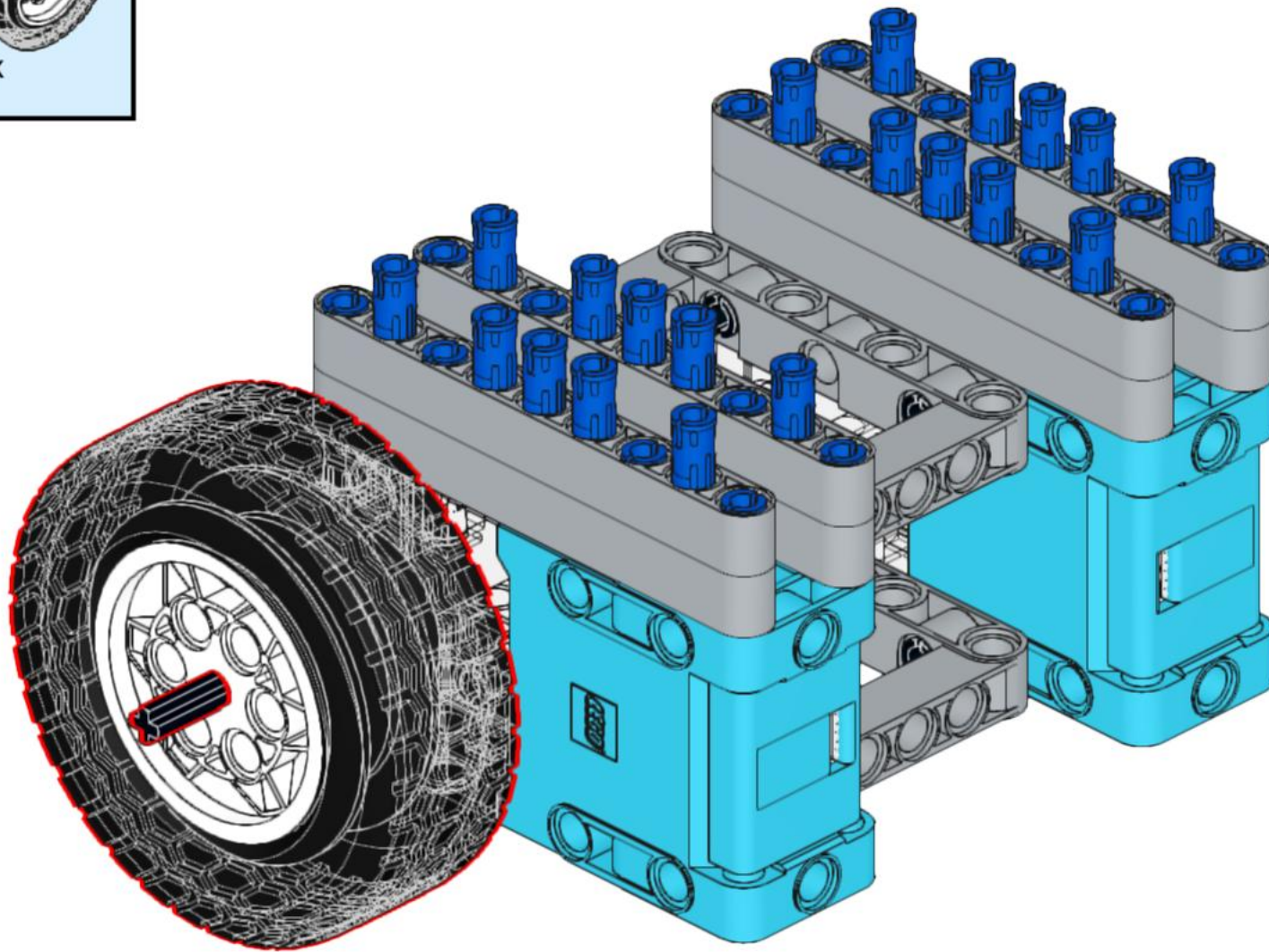
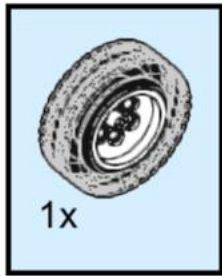
19



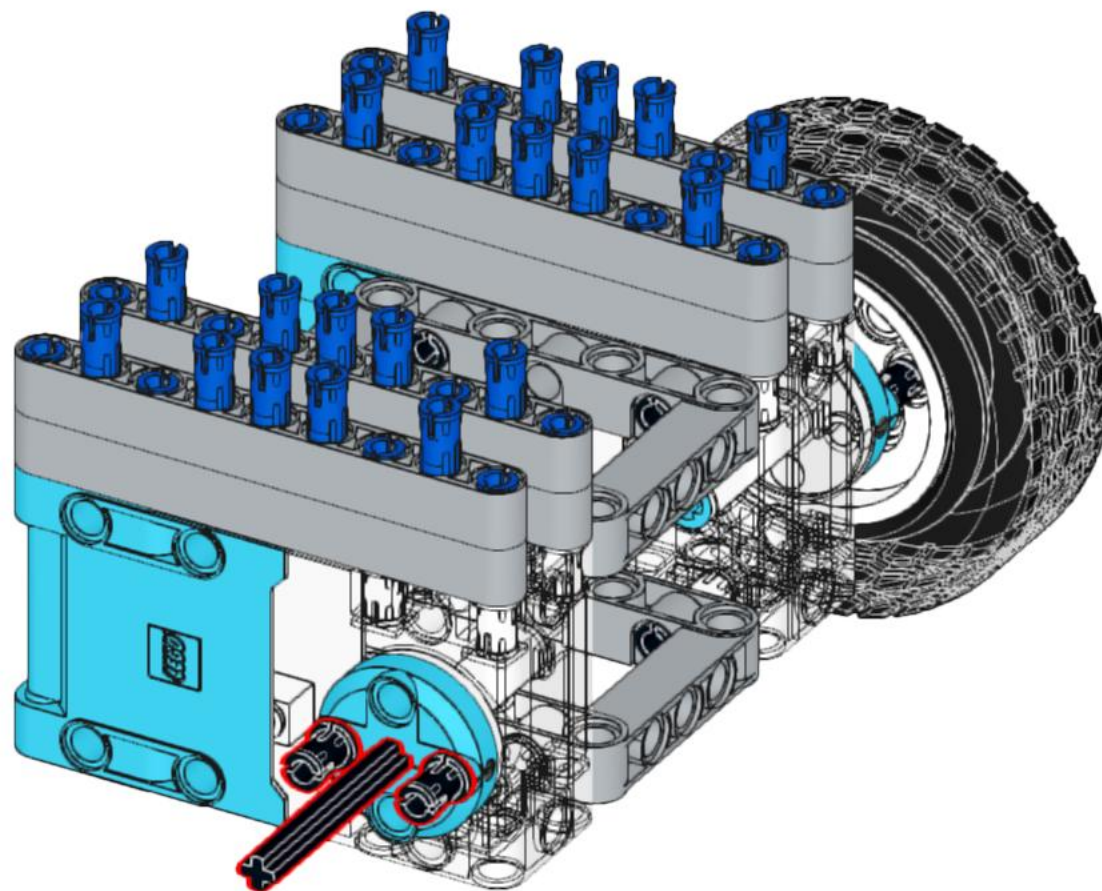
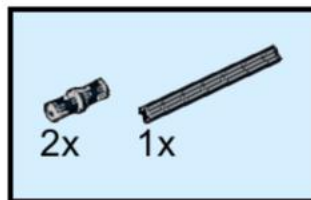
20



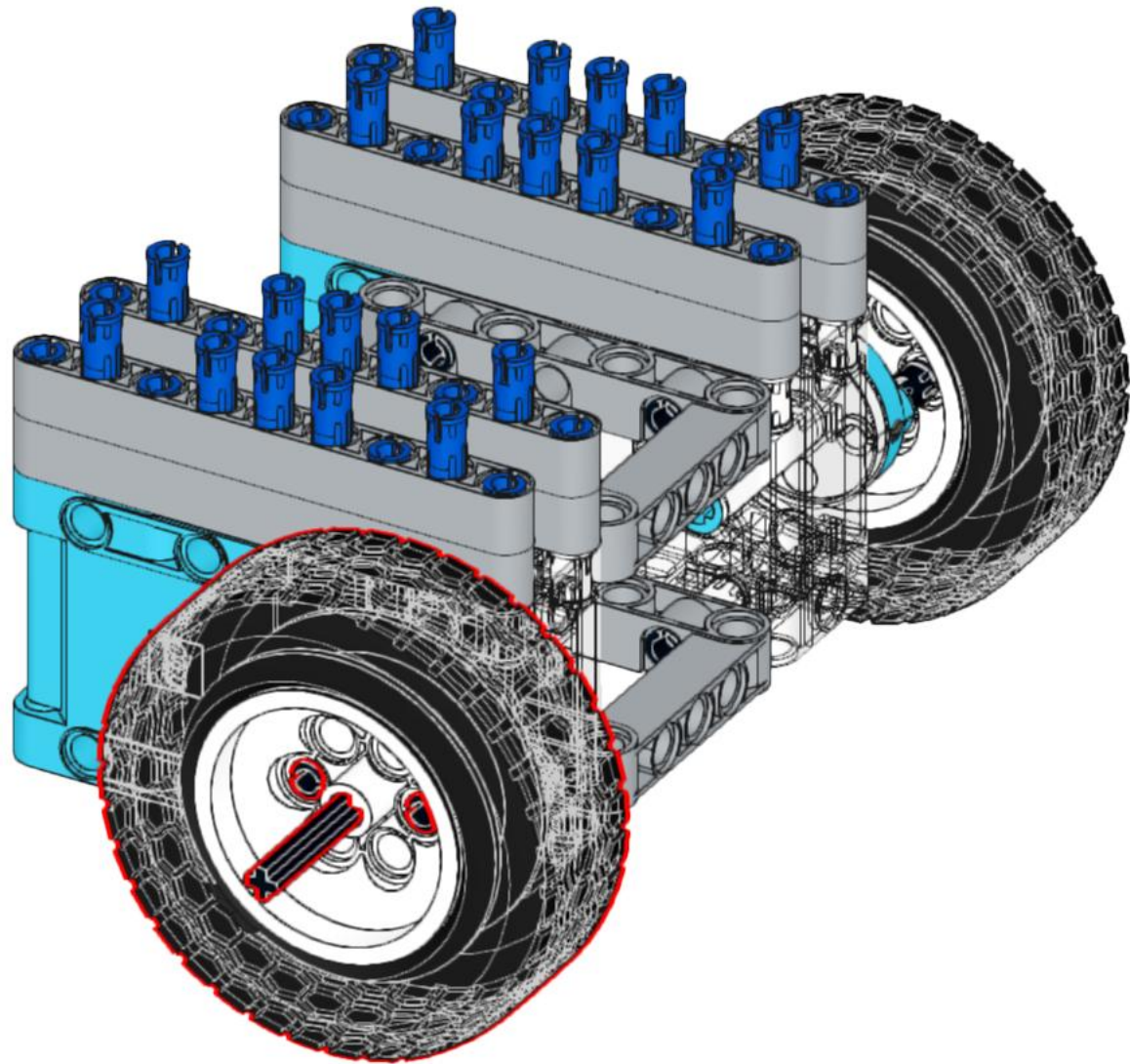
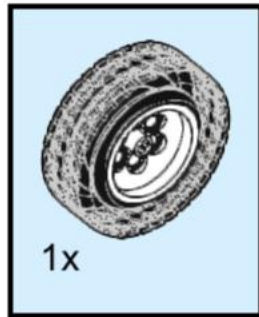
21



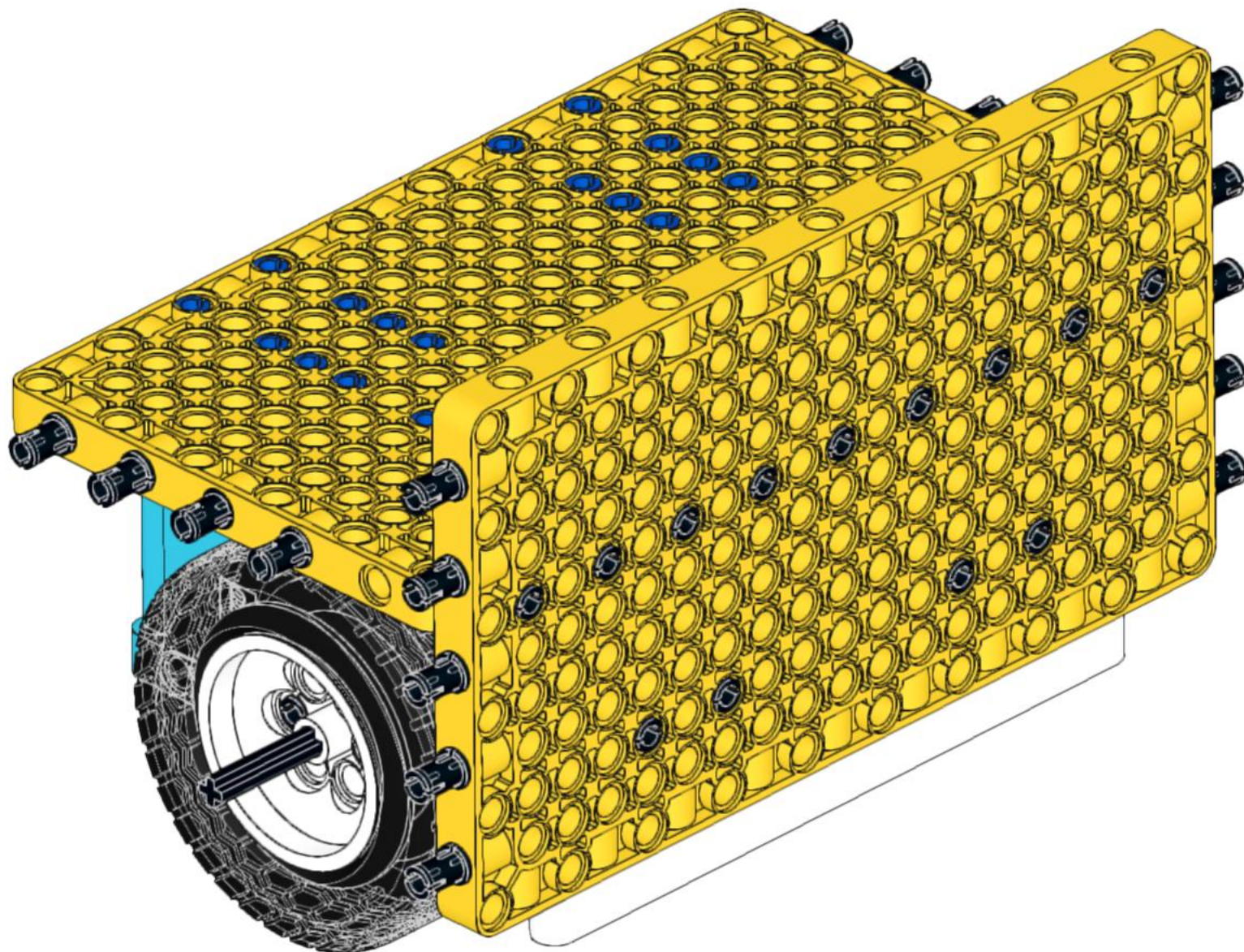
22



23



24

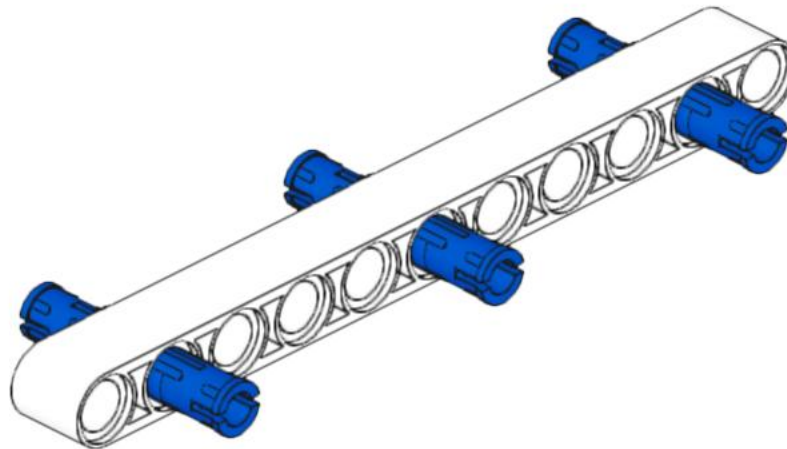
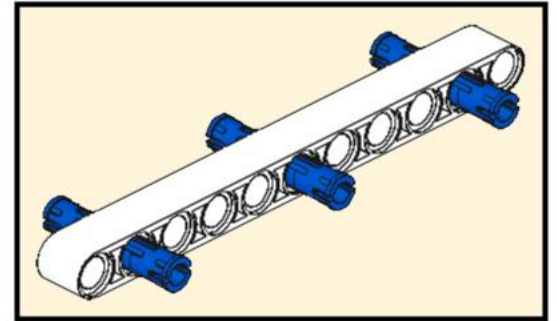
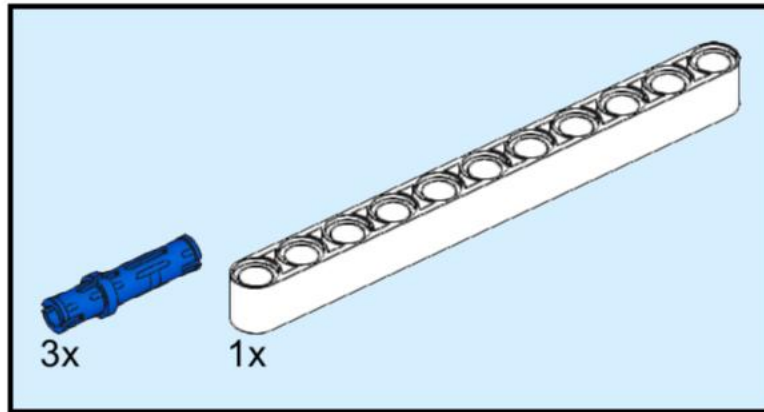


Rear Assembly

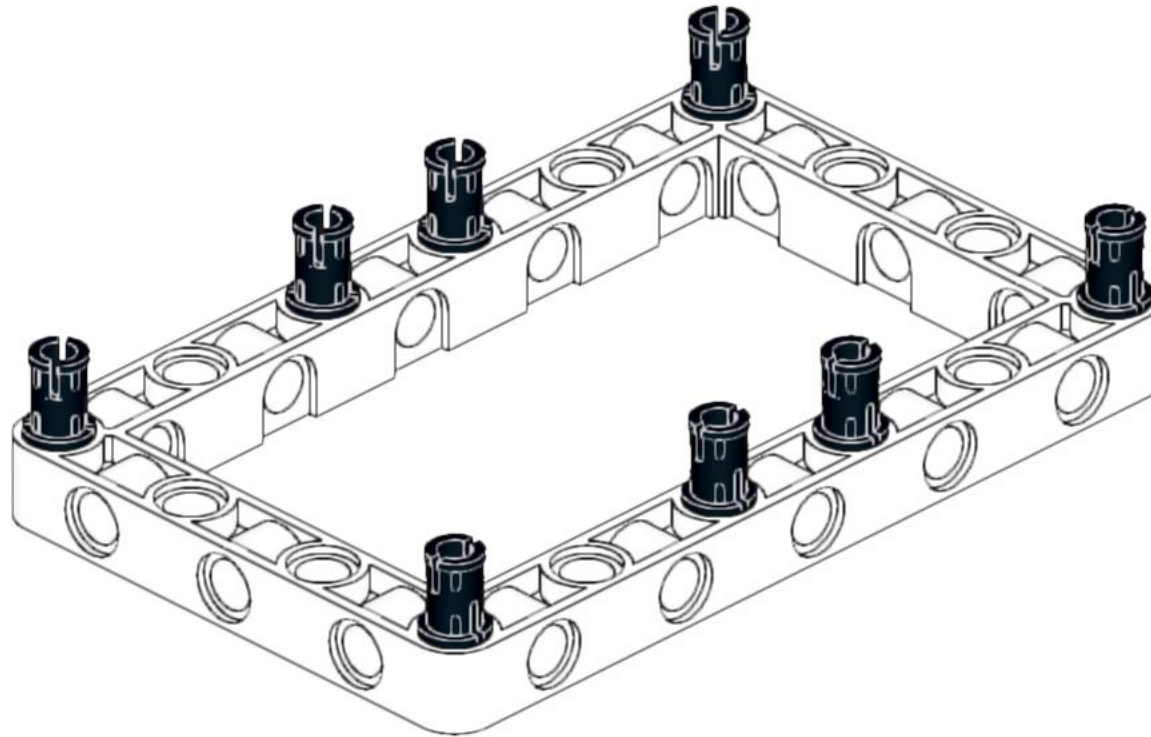
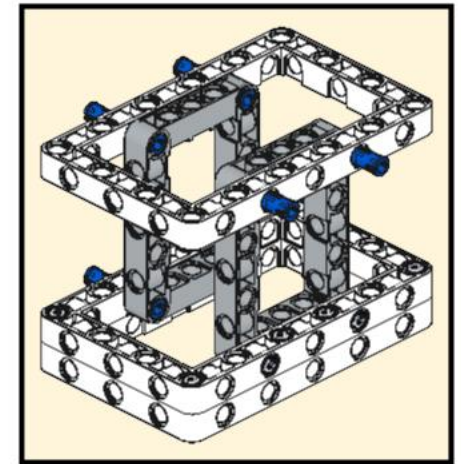
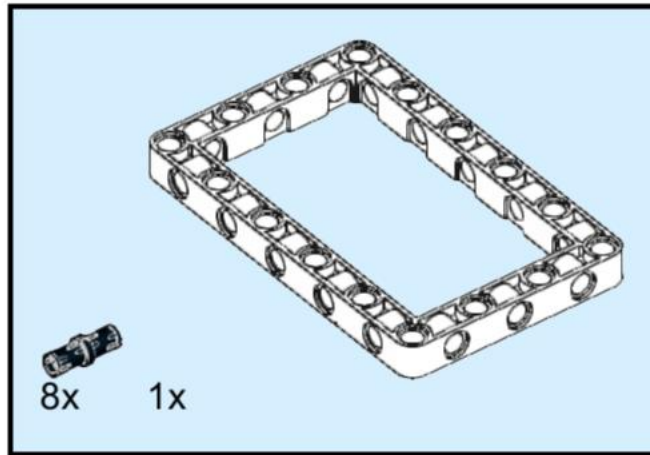
This is the portion of the robot that contains the hub.

While not shown in these plans, we sometimes installed two 2×6×2 Brick, Counter Weights (73090) in the middle 5x7 frame to offset the weight of the front of the robot.

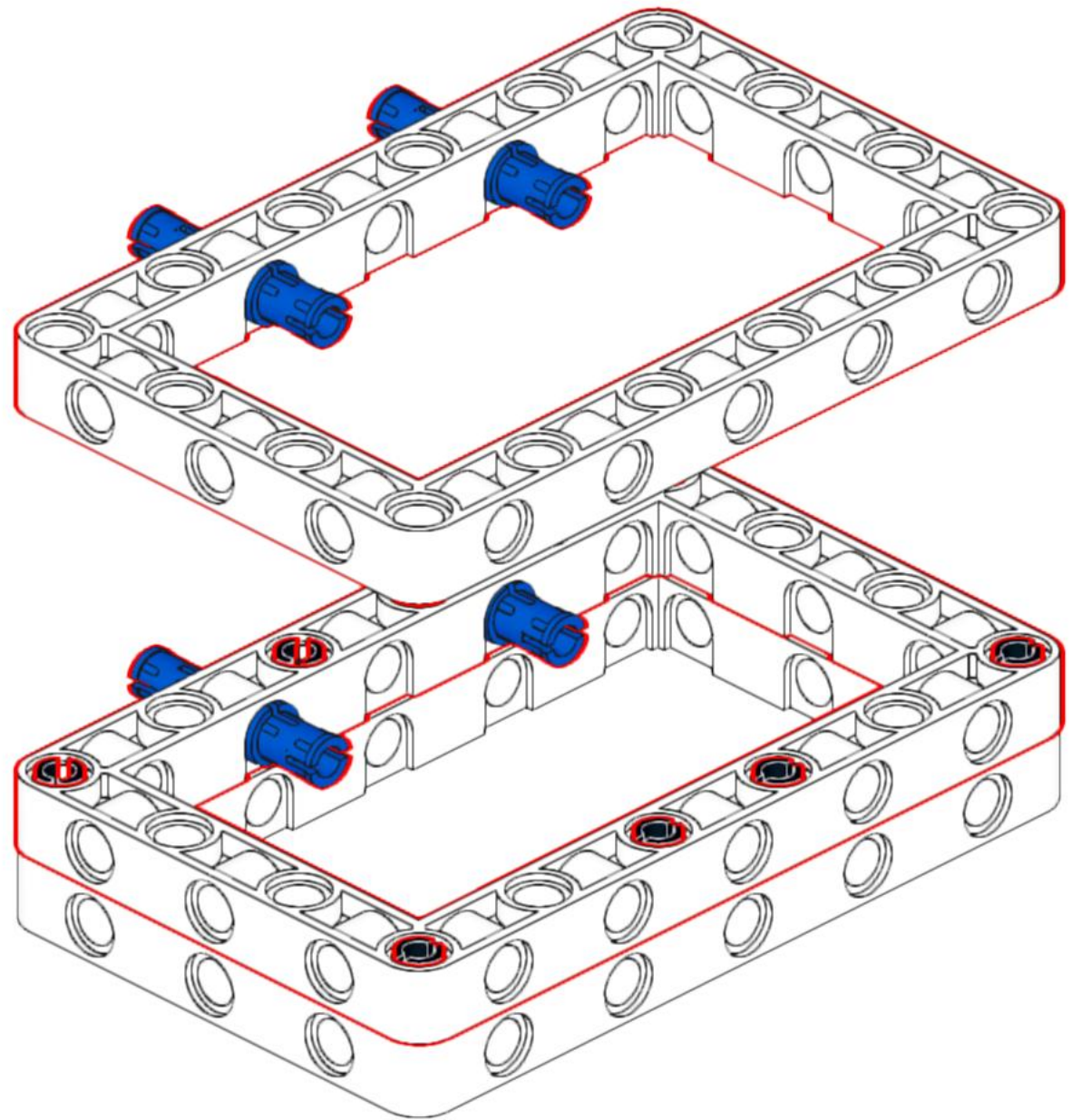
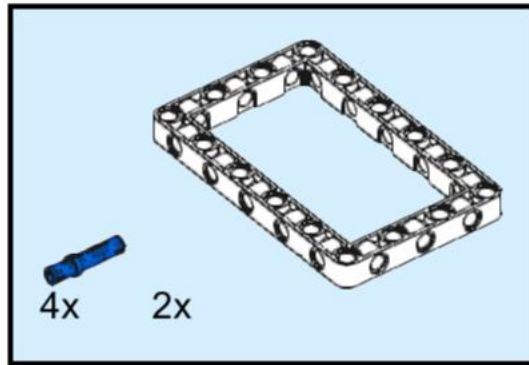
25



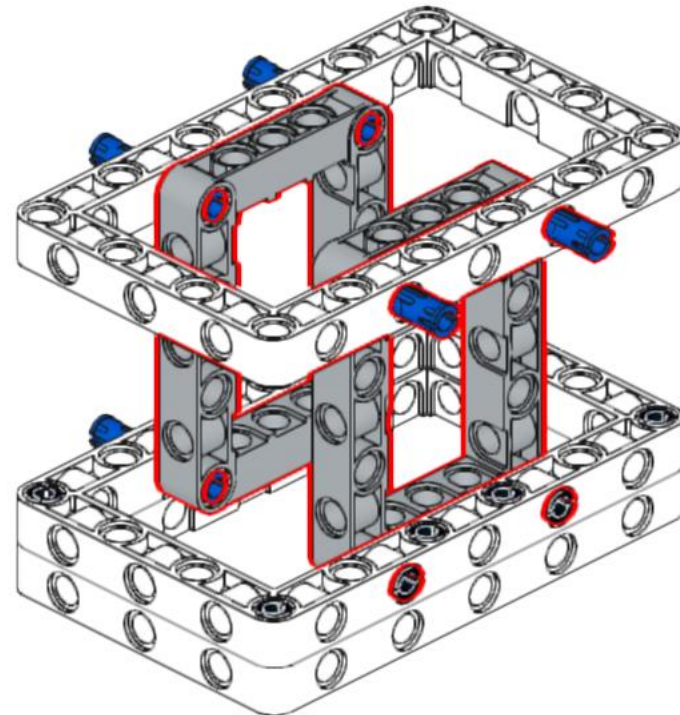
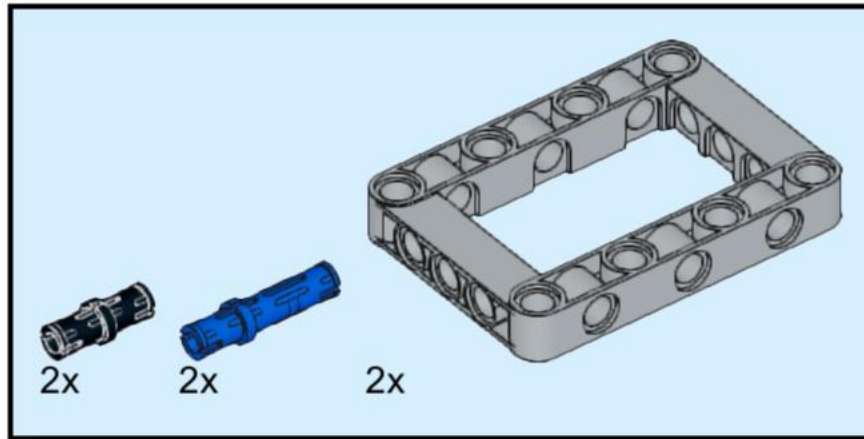
26



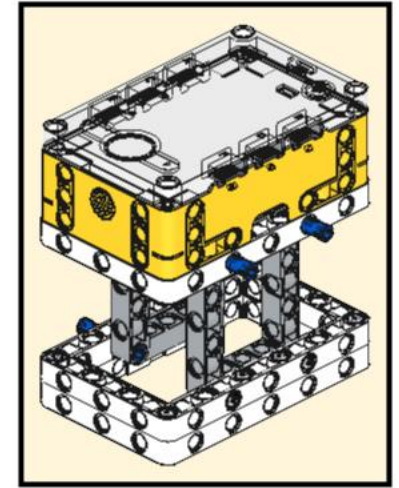
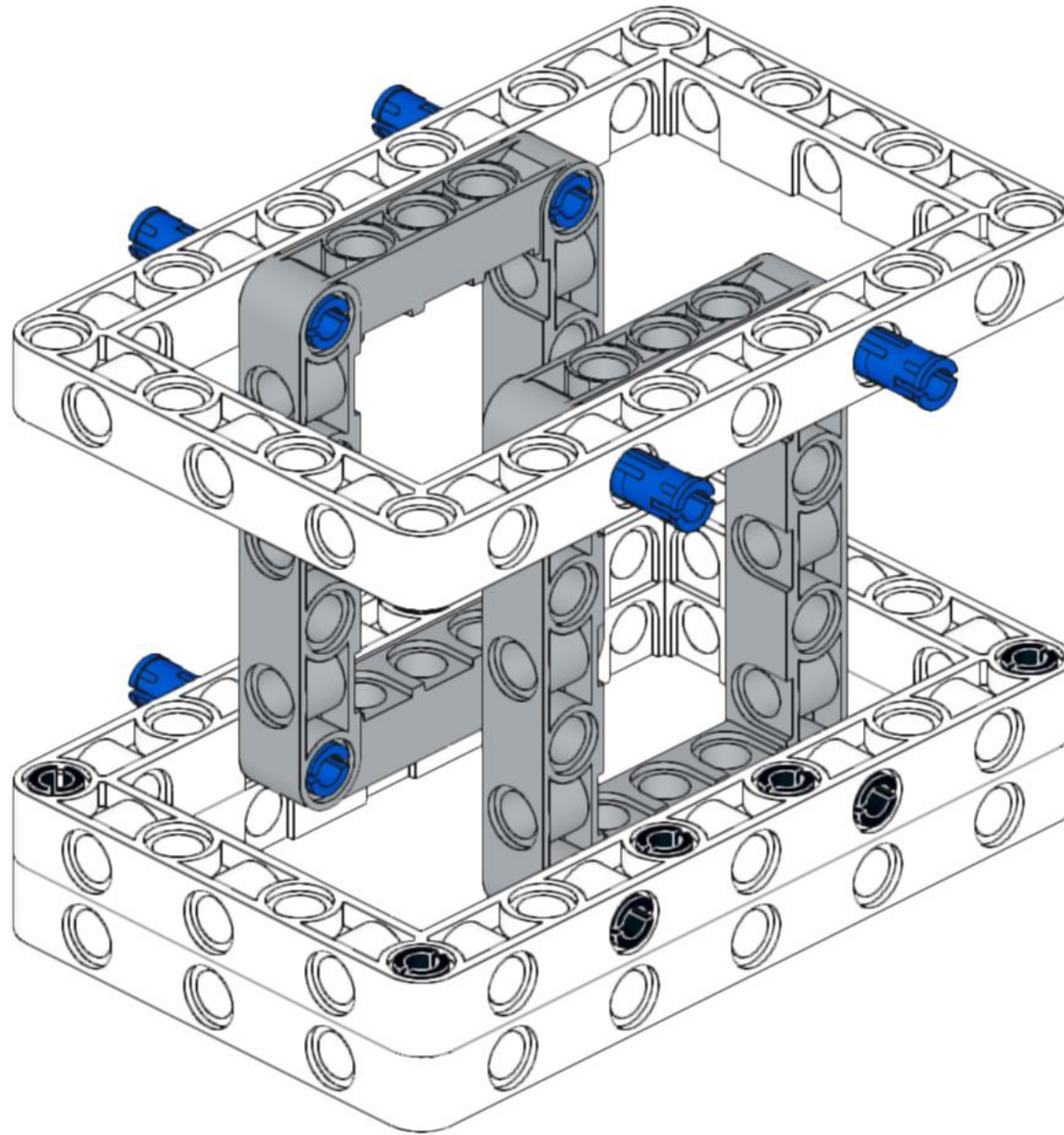
27



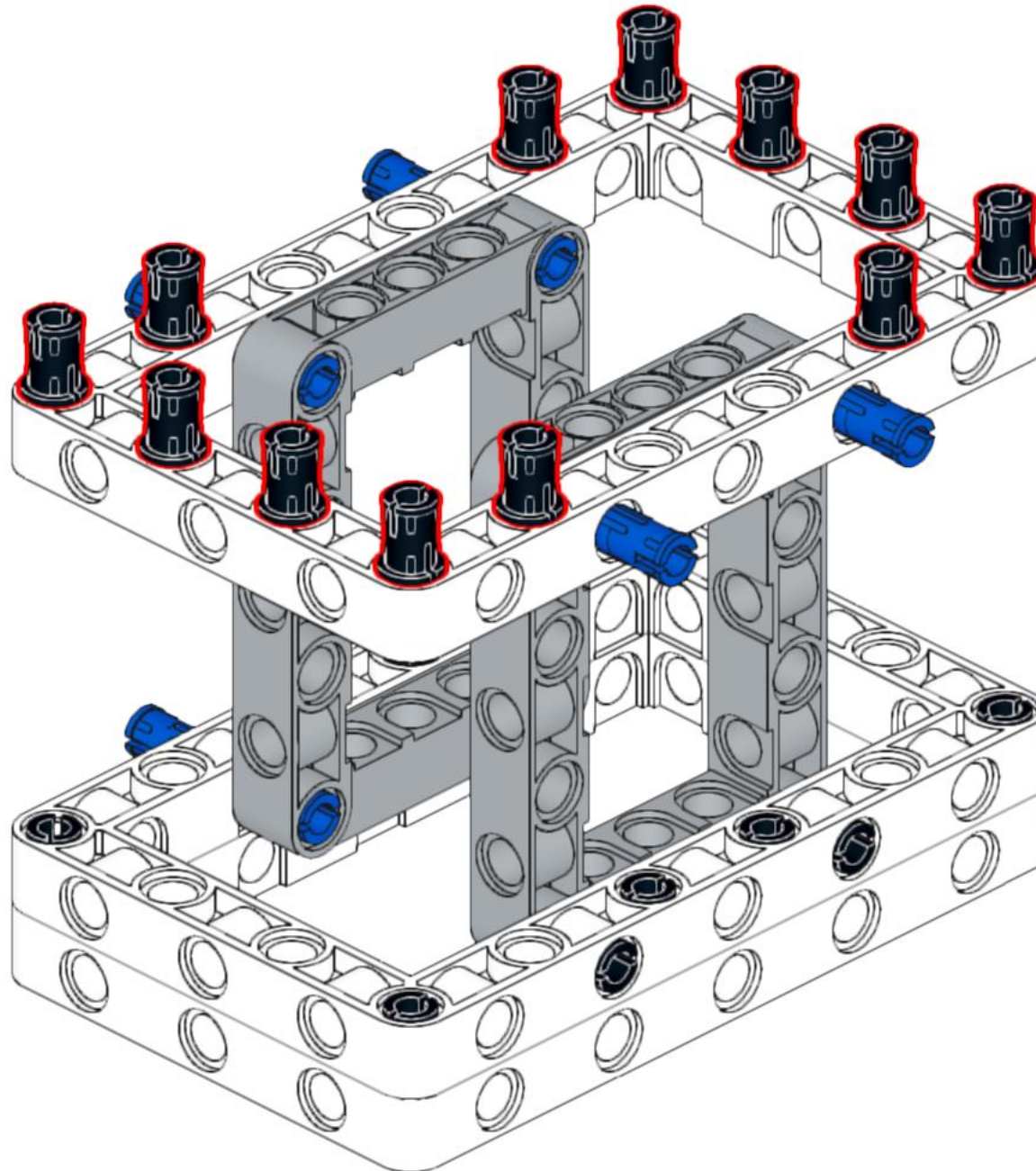
28



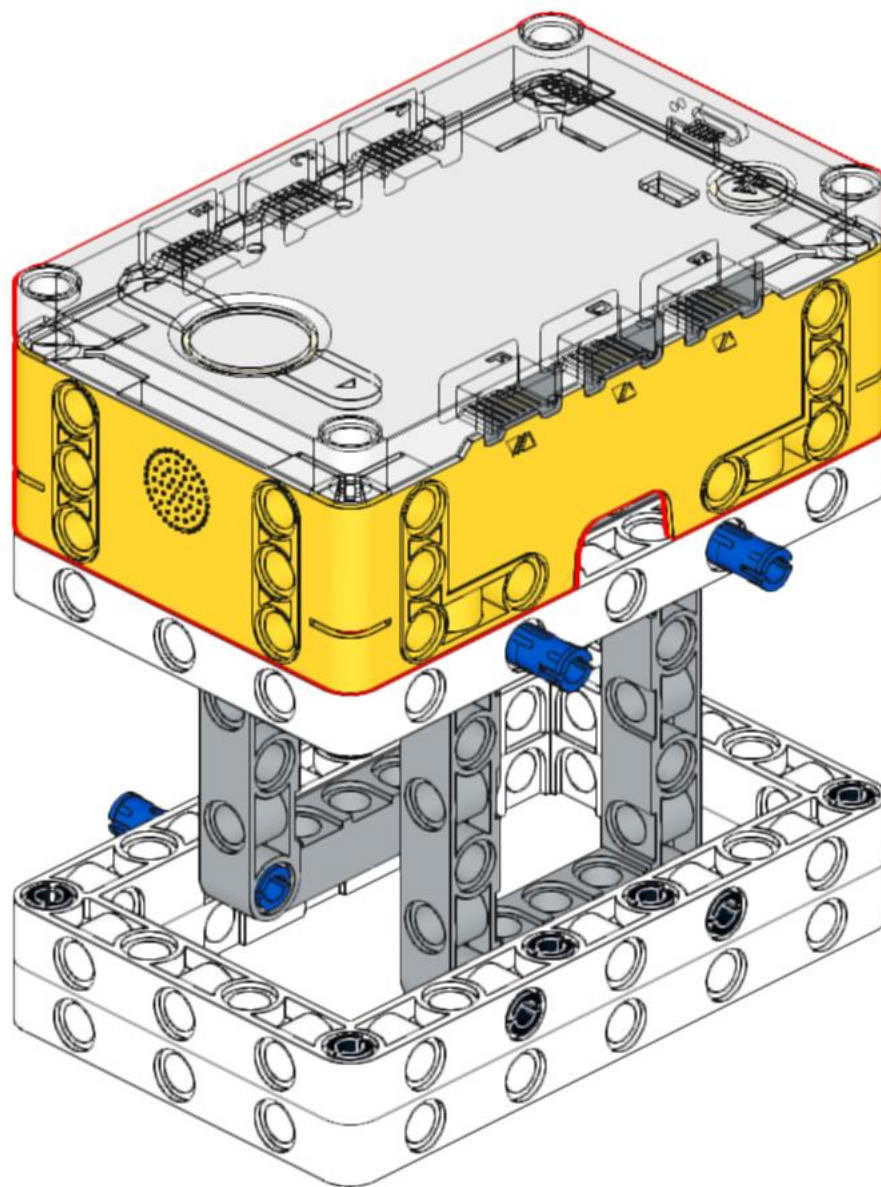
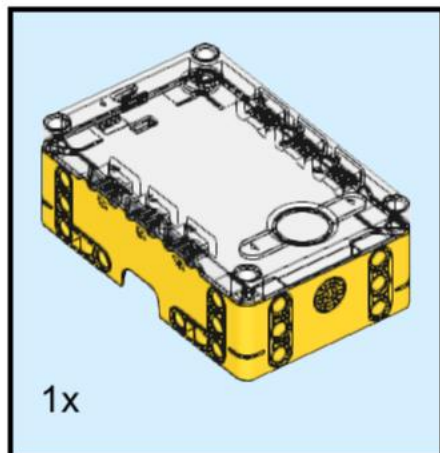
29



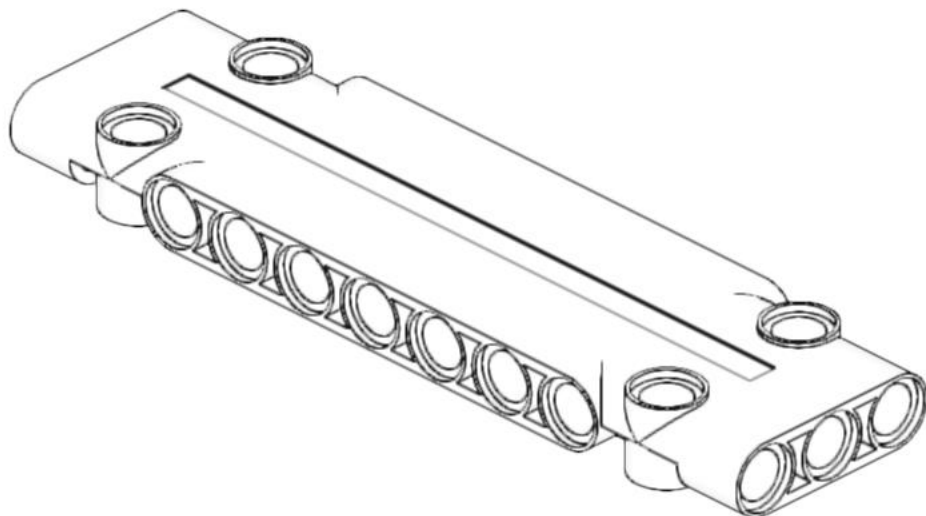
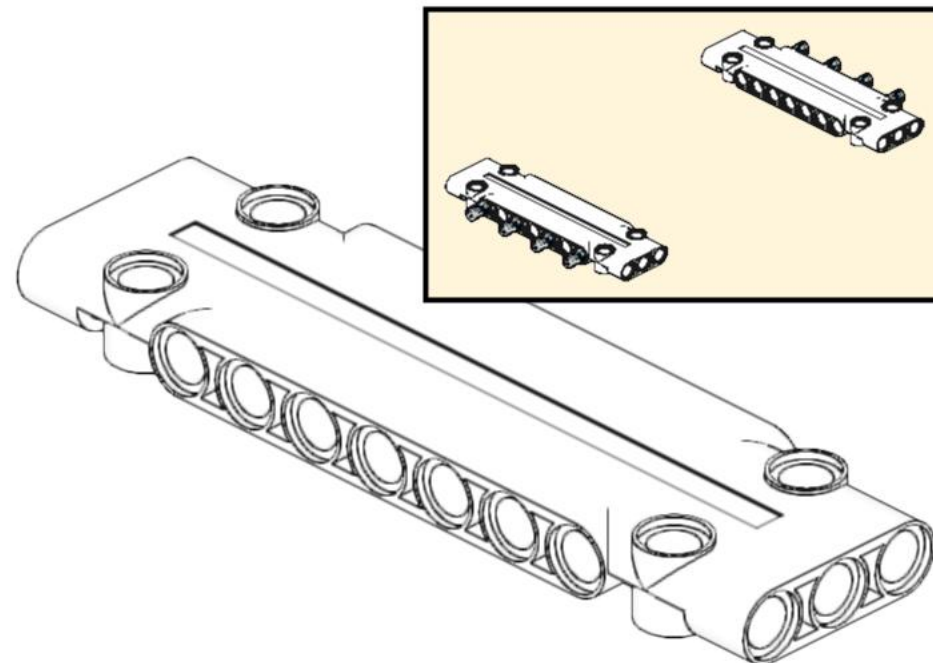
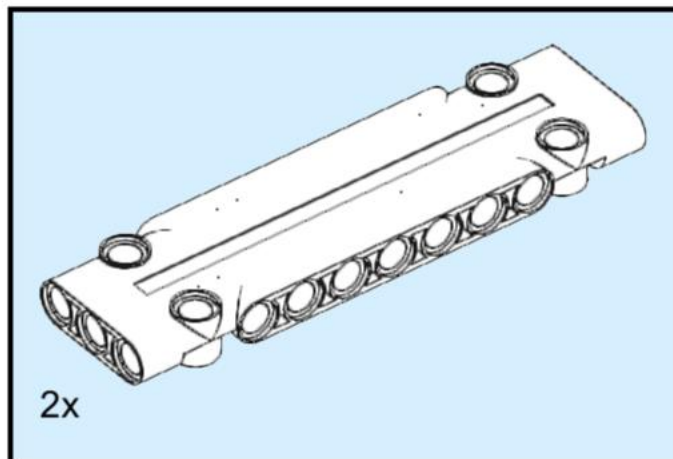
30



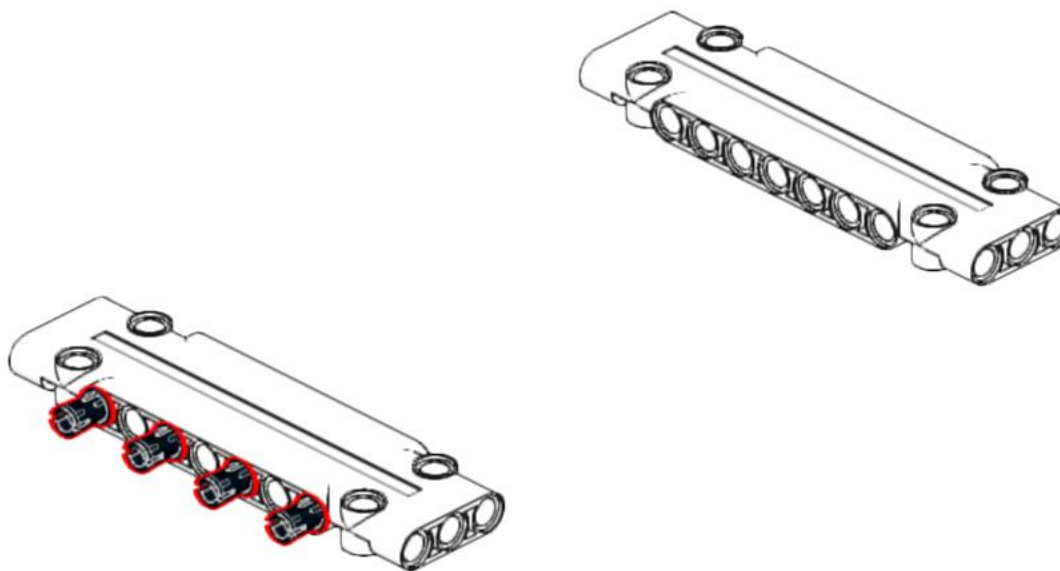
31



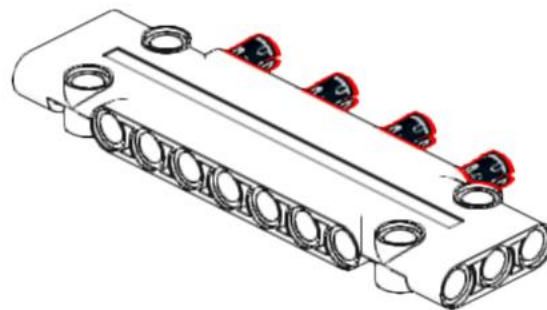
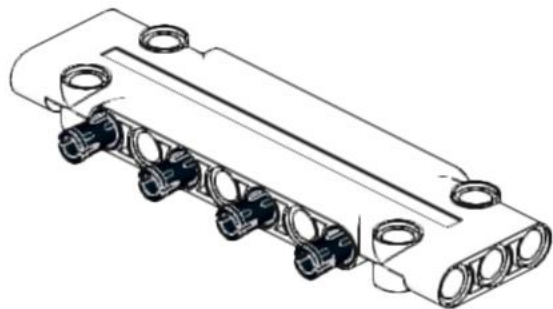
32



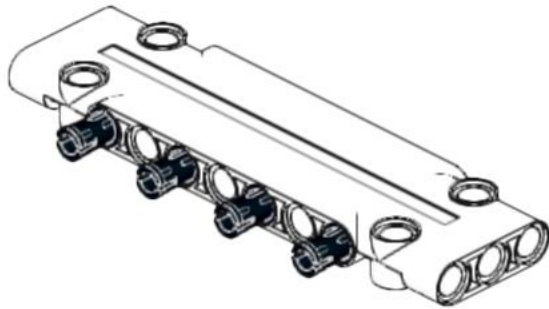
33



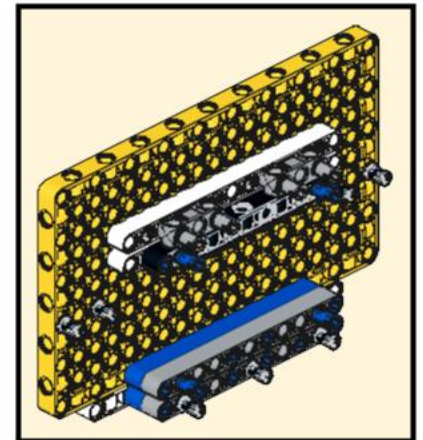
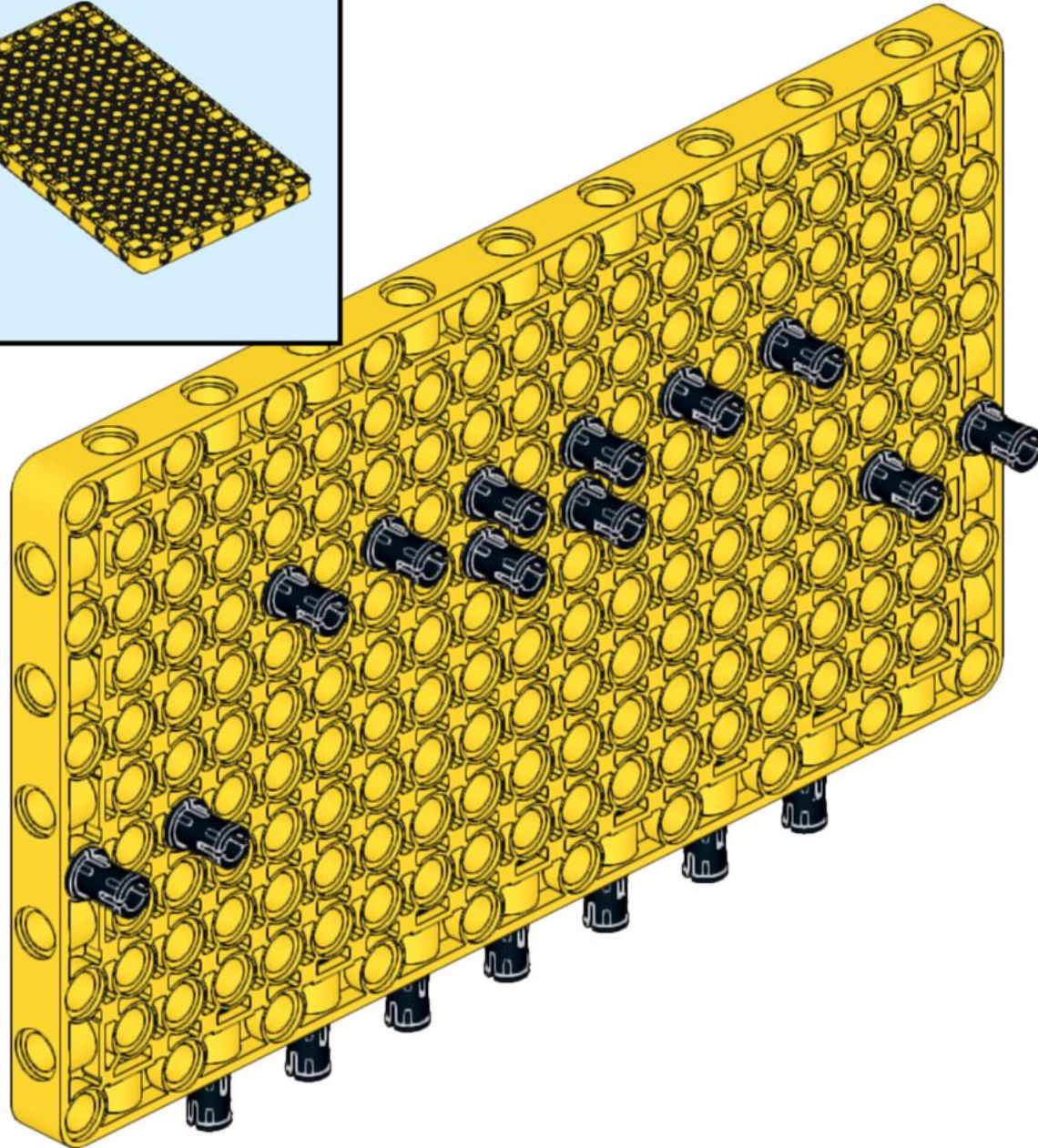
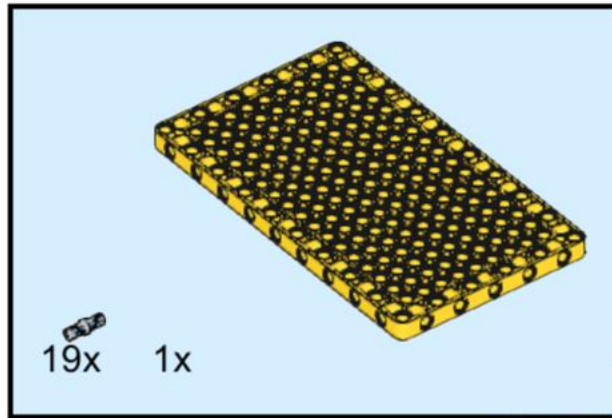
34



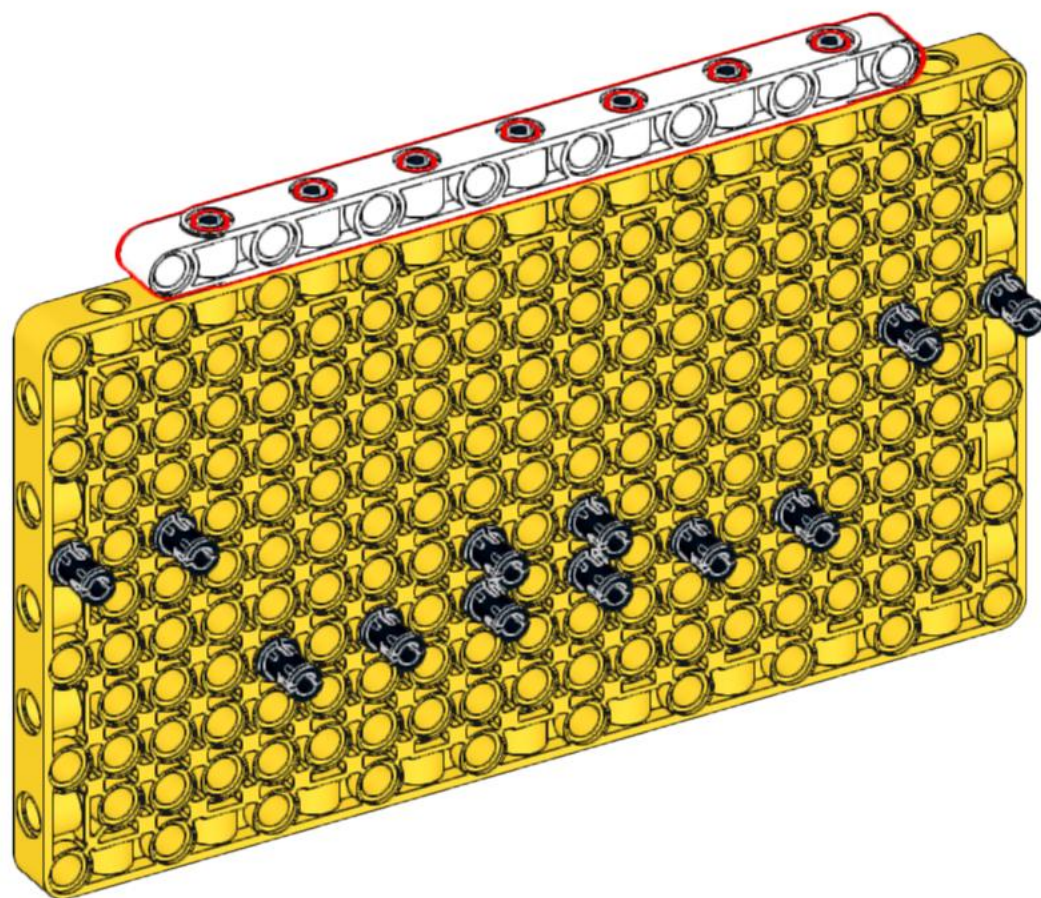
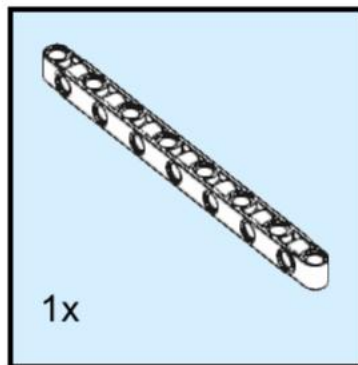
35



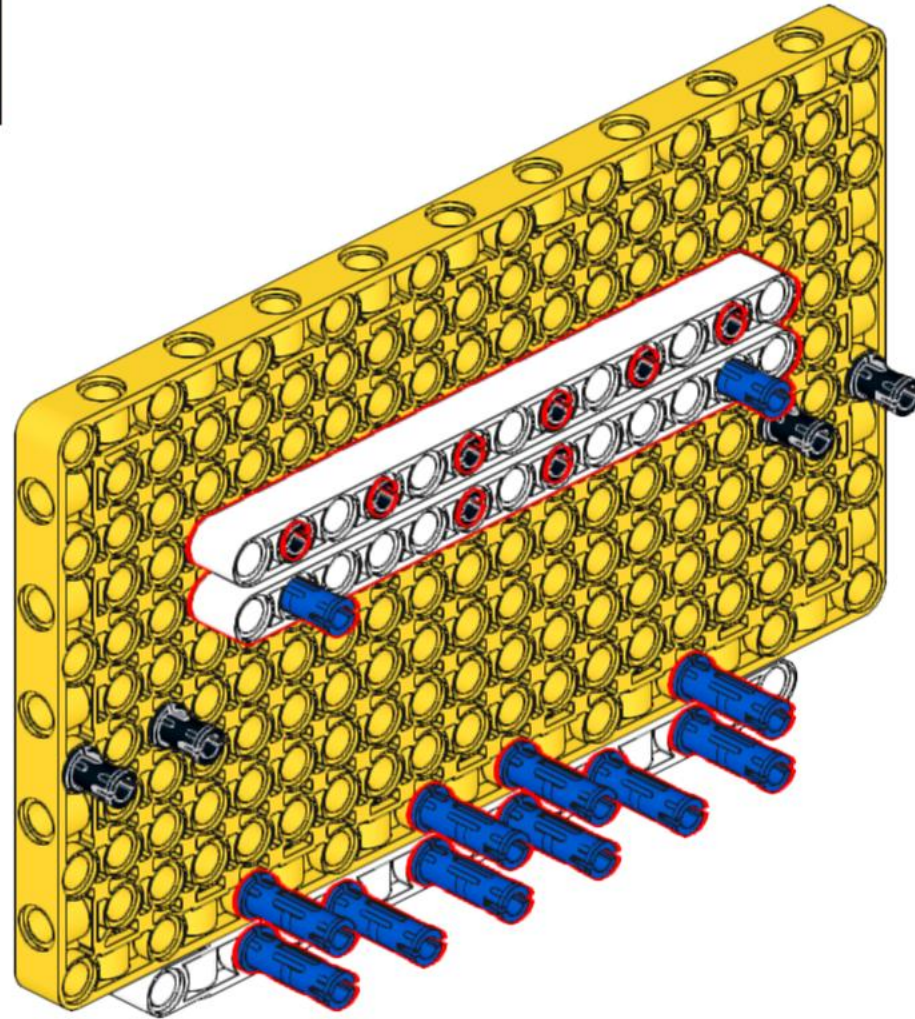
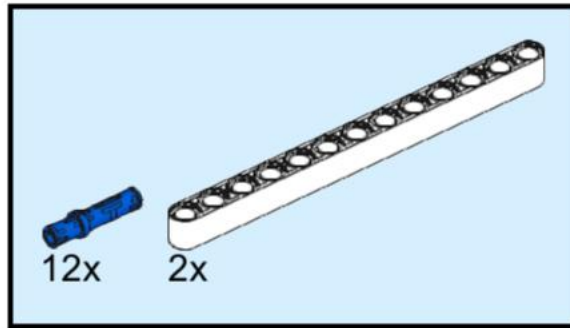
36



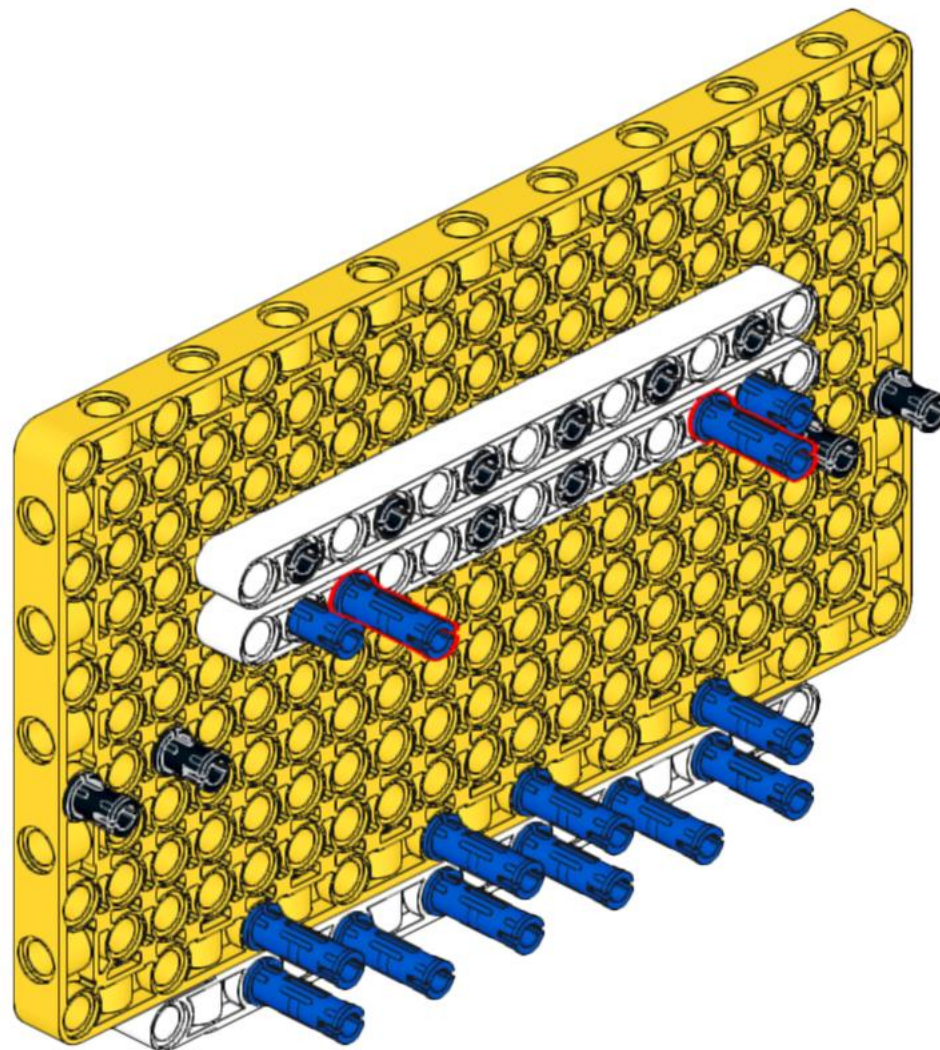
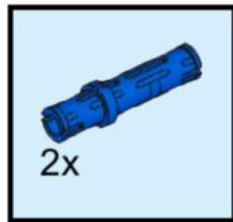
37



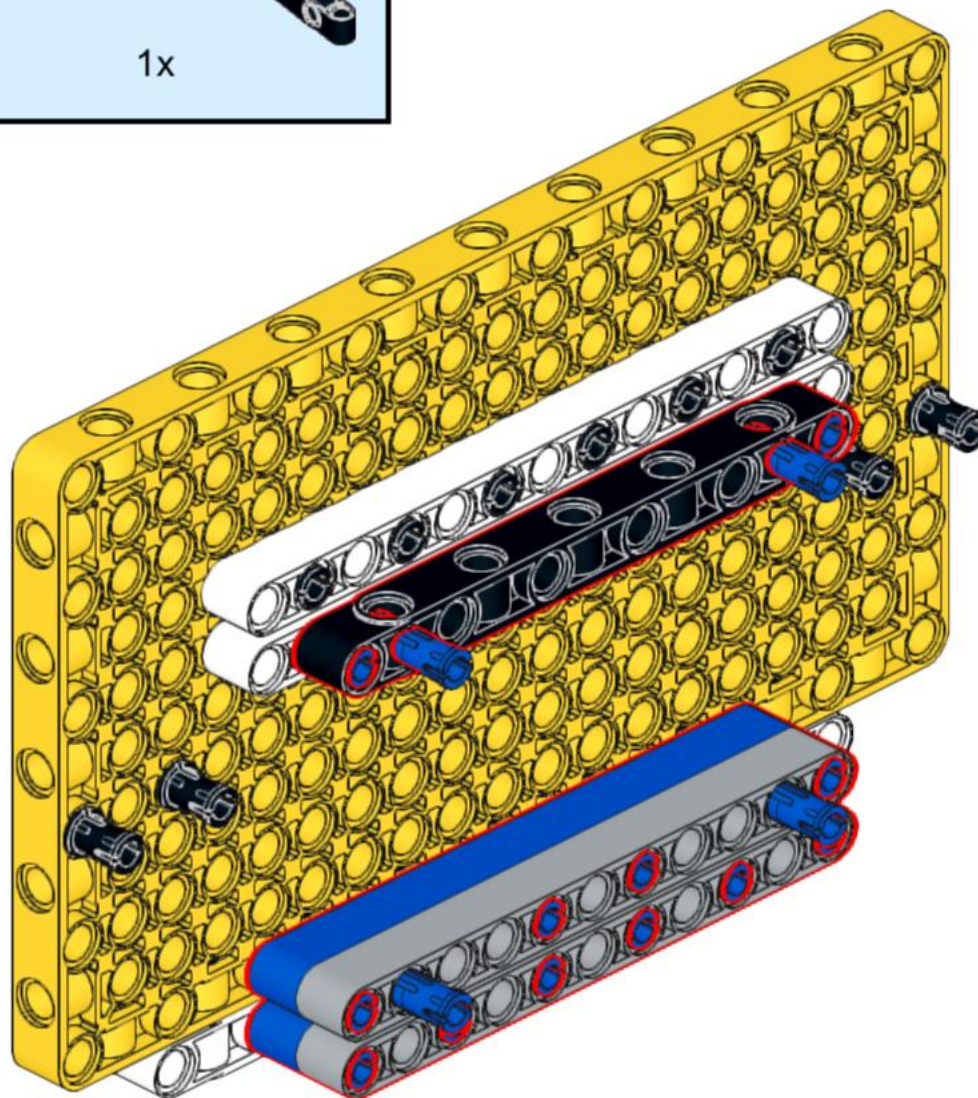
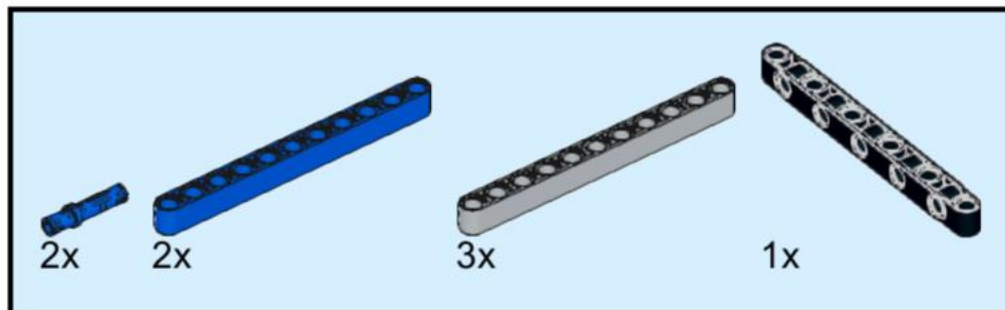
38



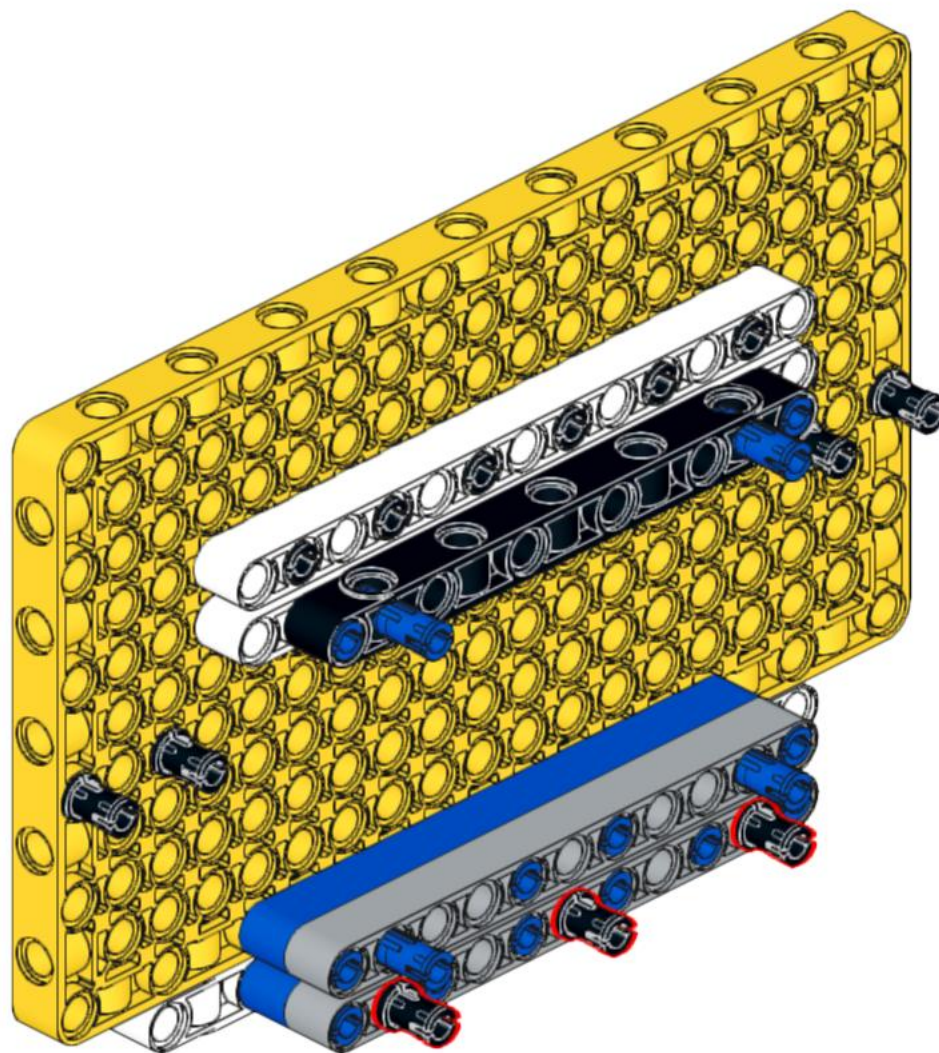
39



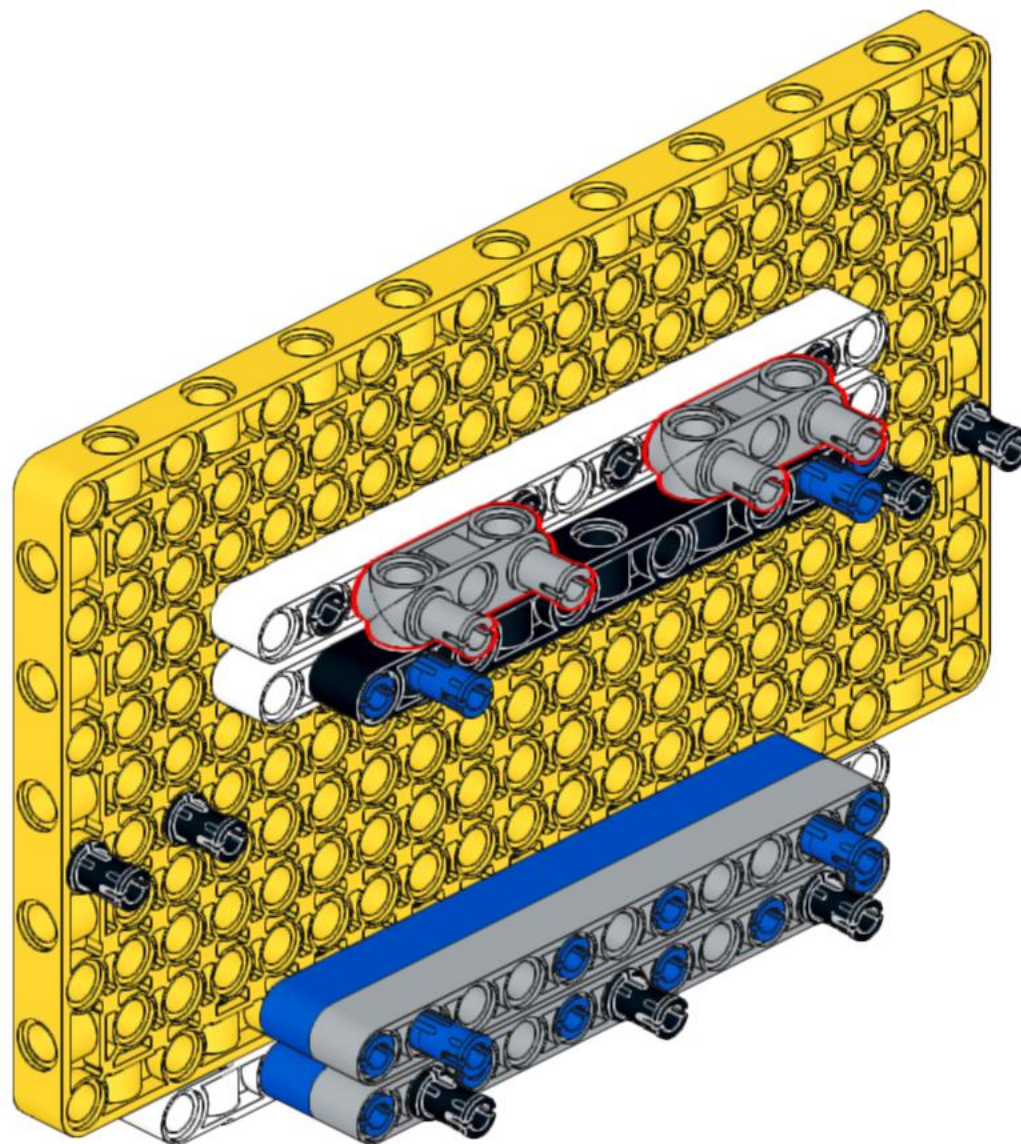
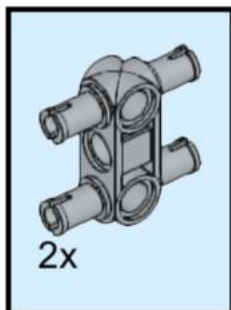
40



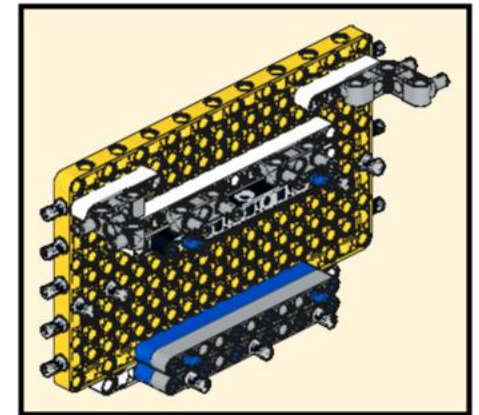
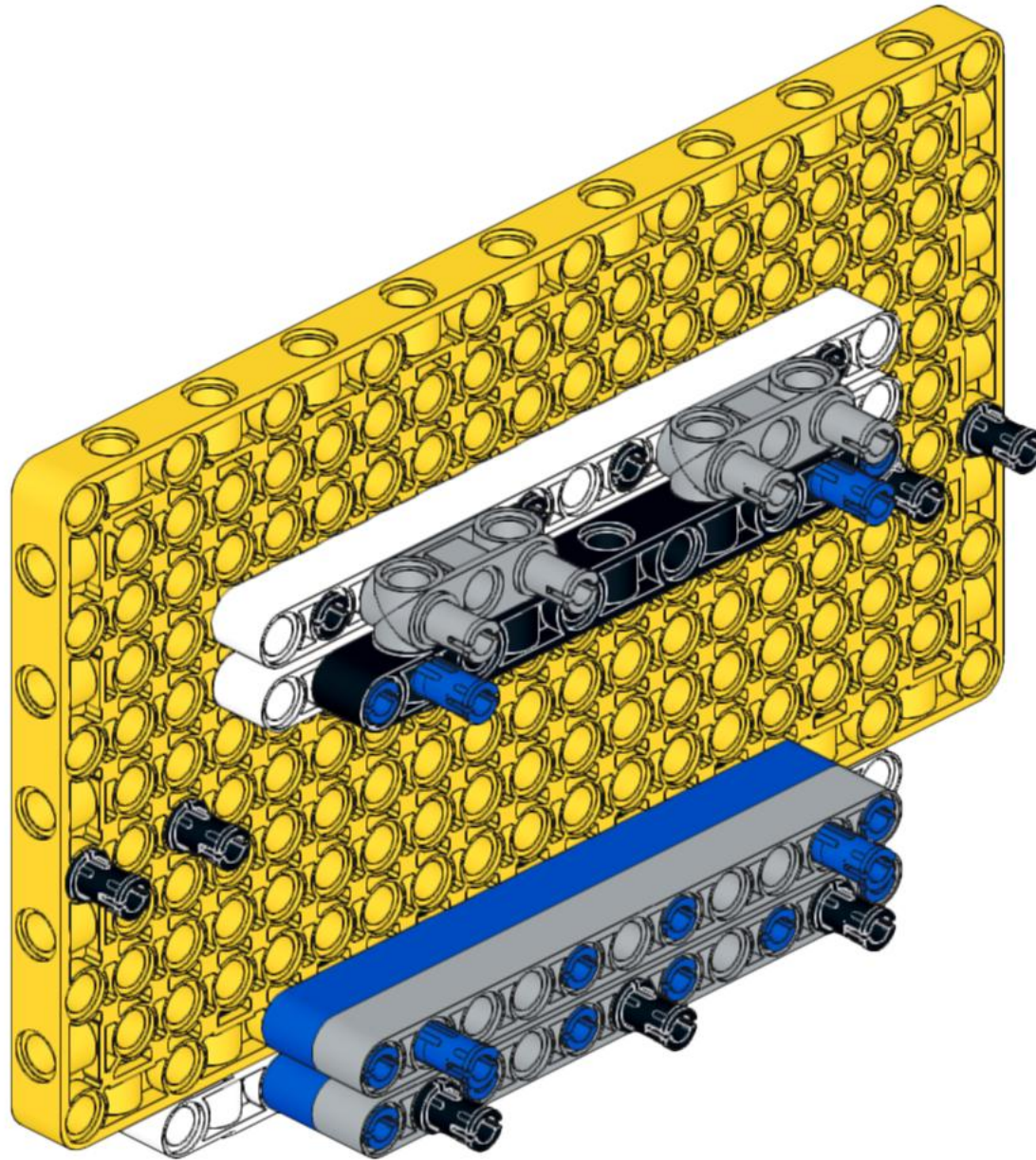
41



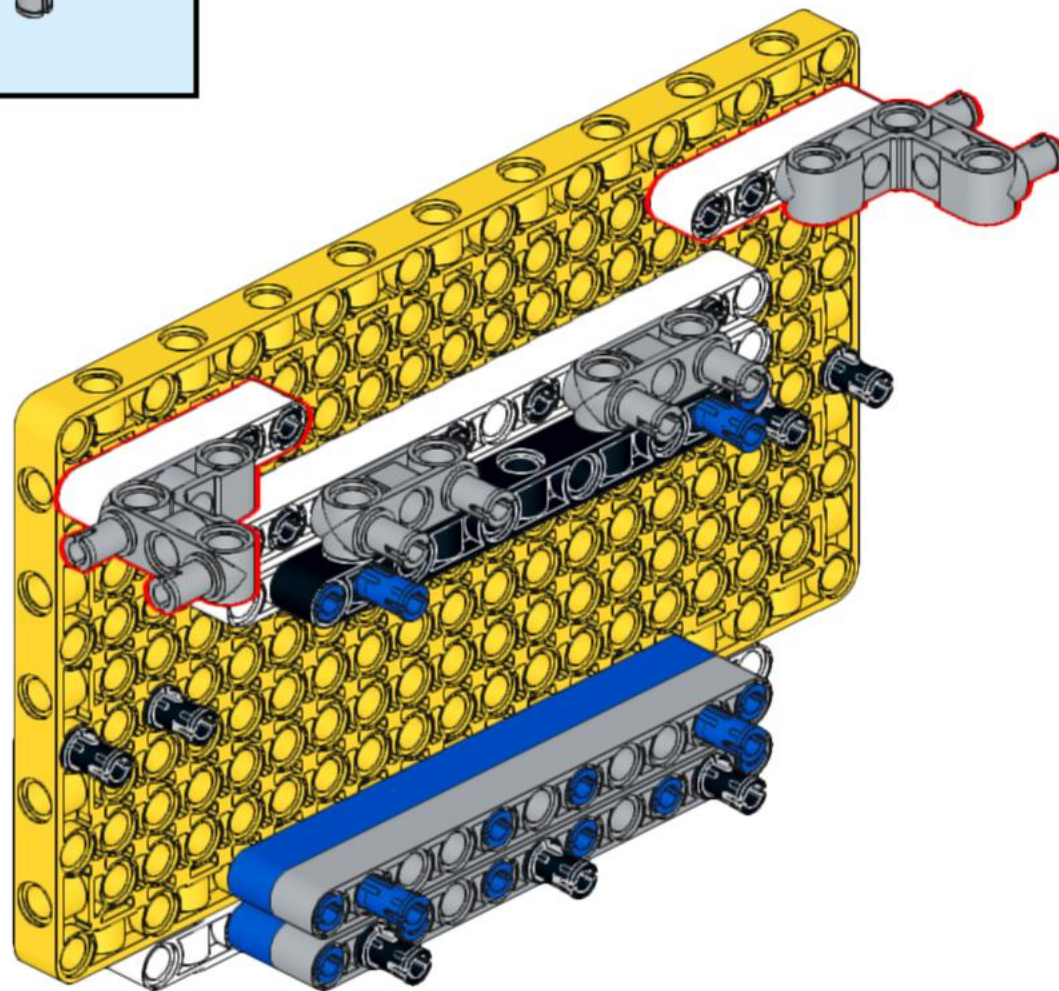
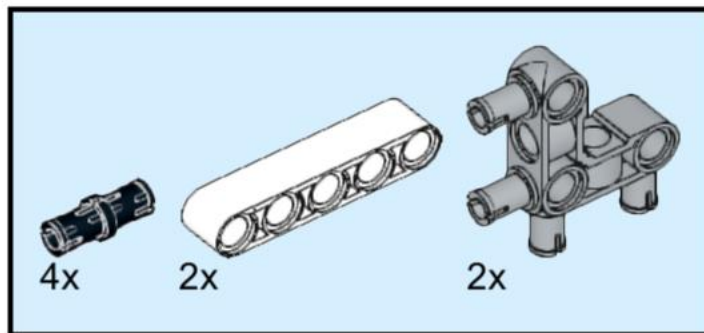
42



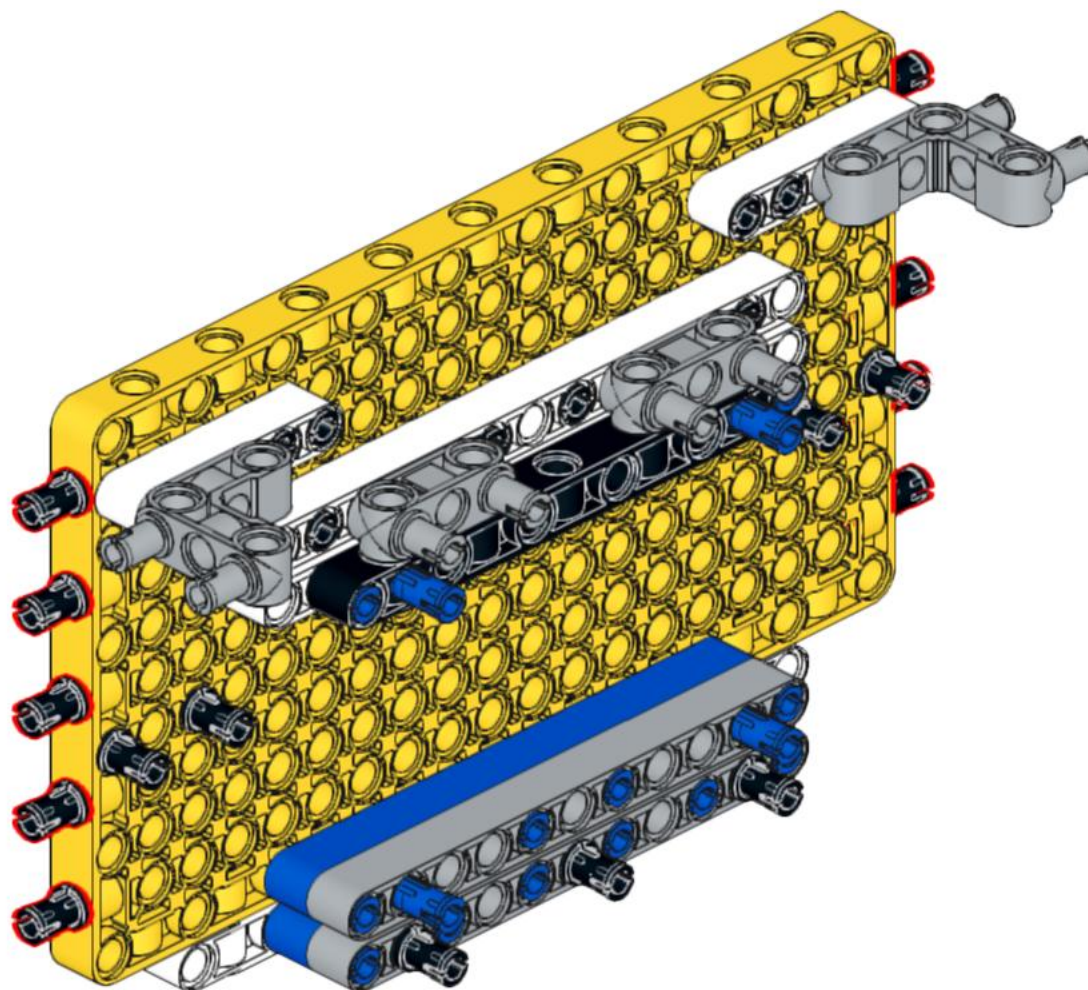
43



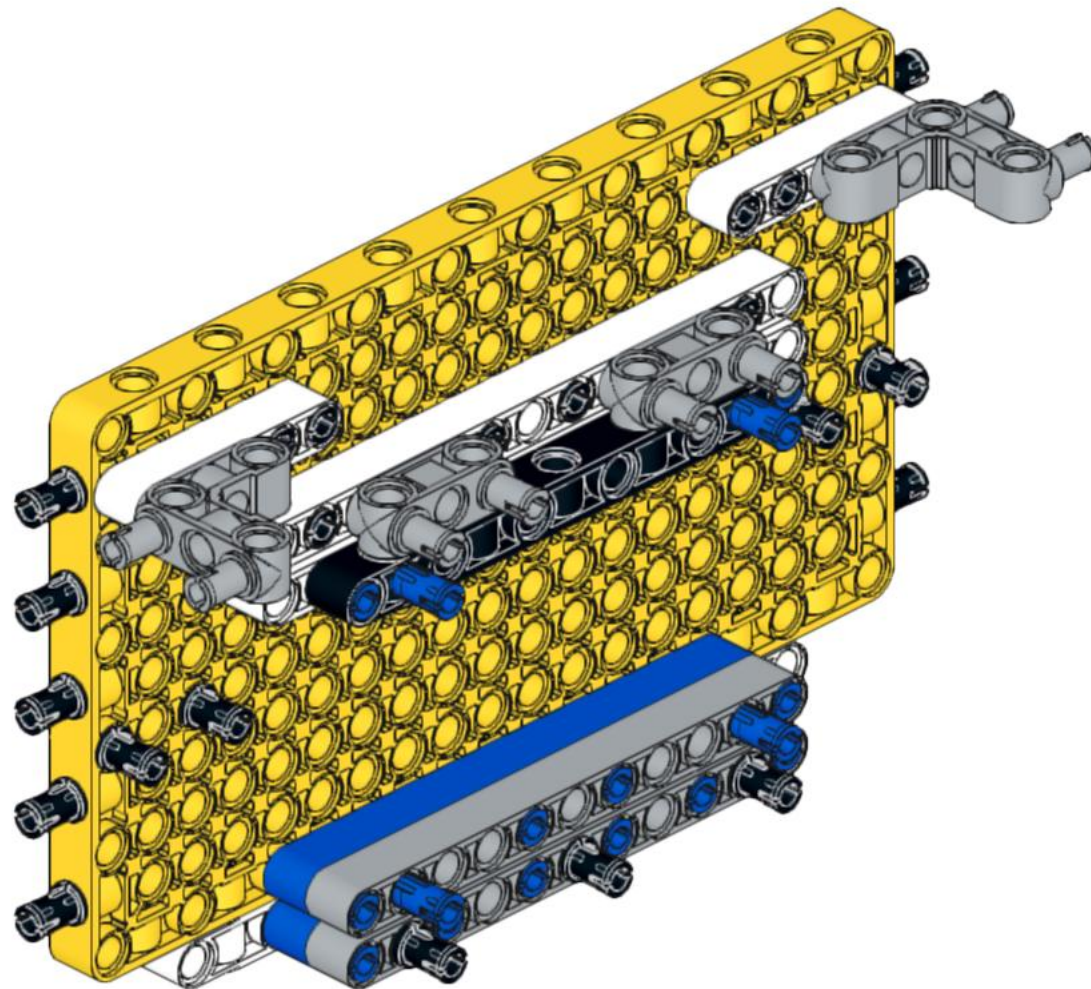
44



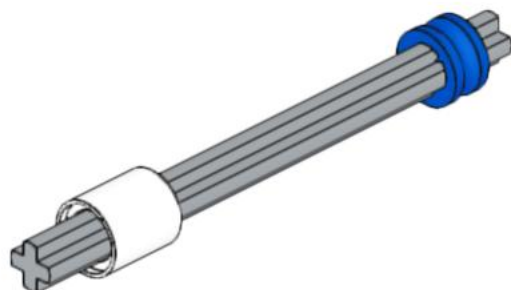
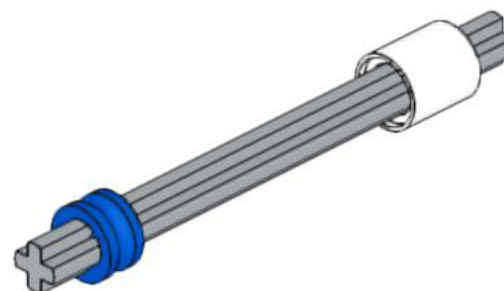
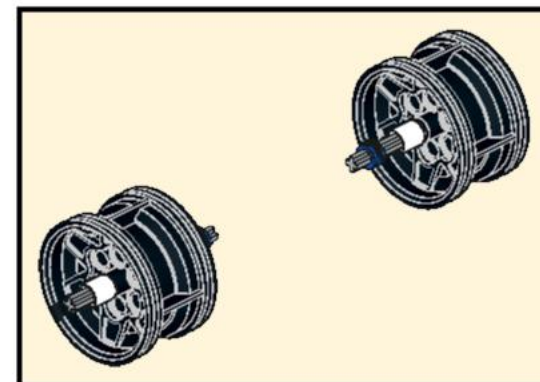
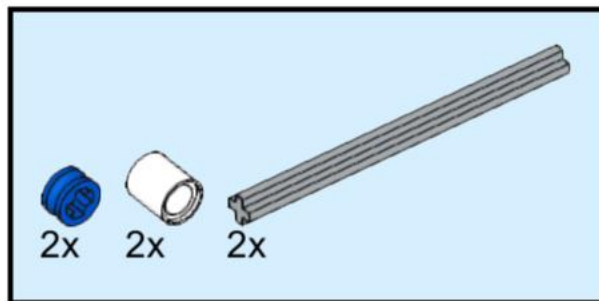
45



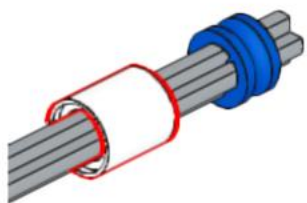
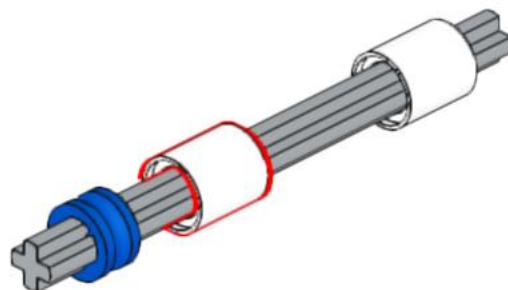
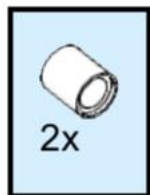
46



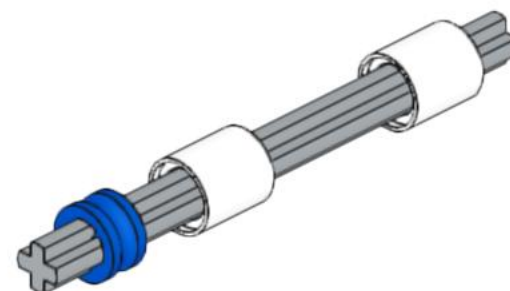
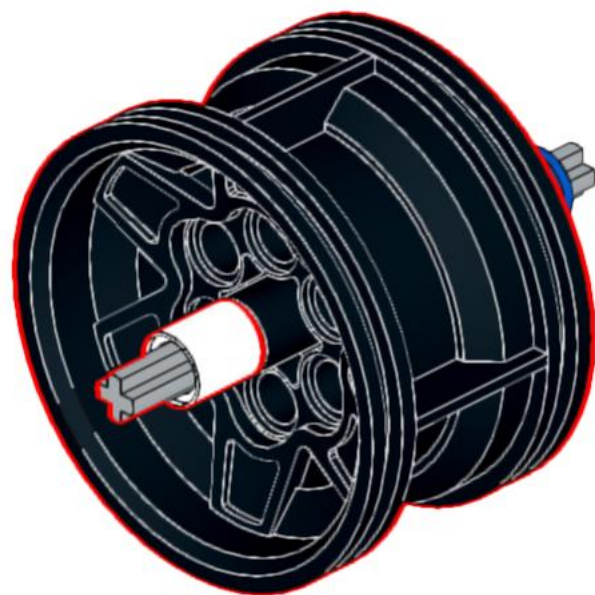
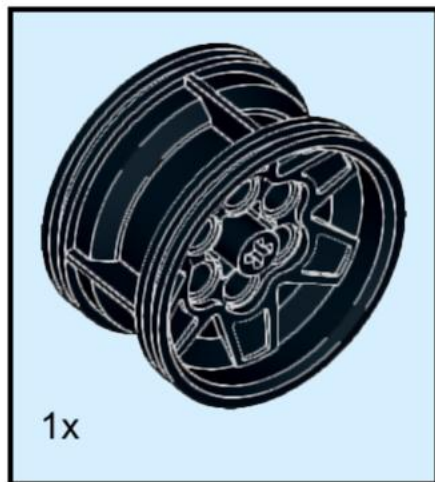
47



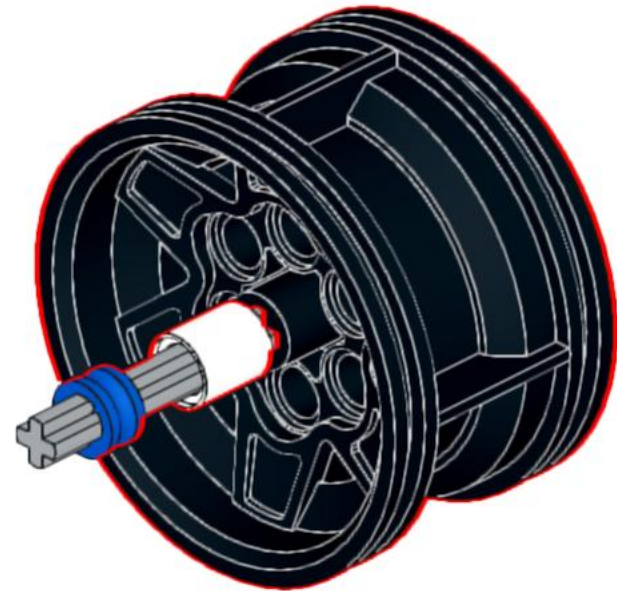
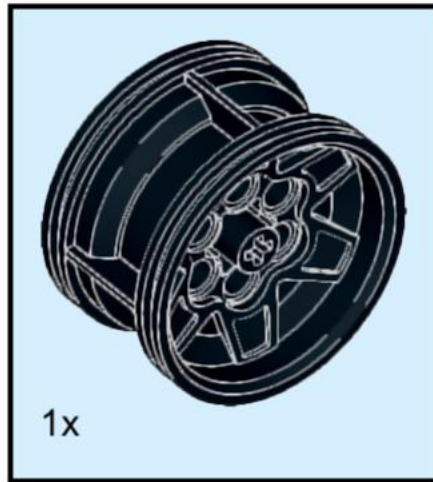
48



49

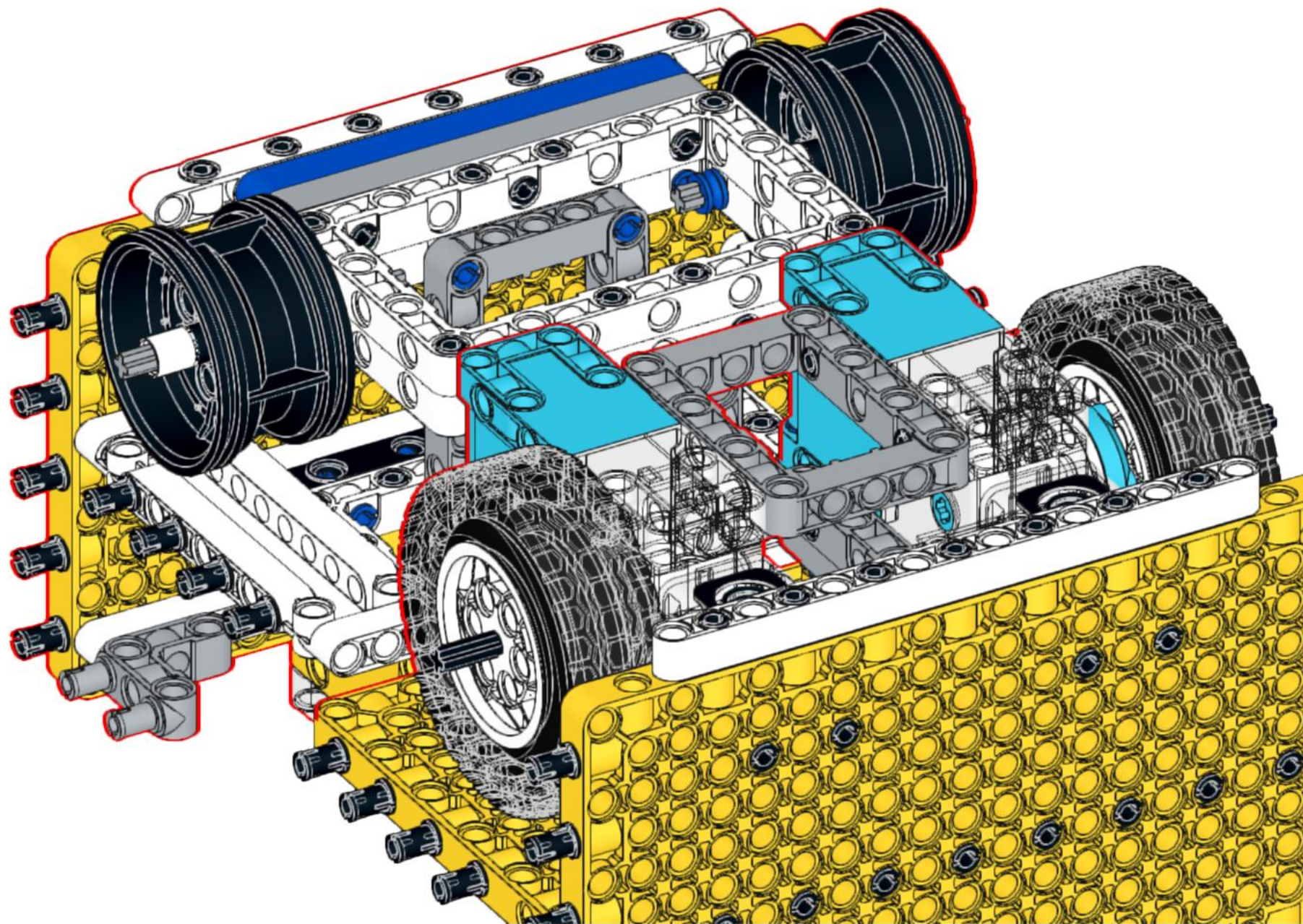


50

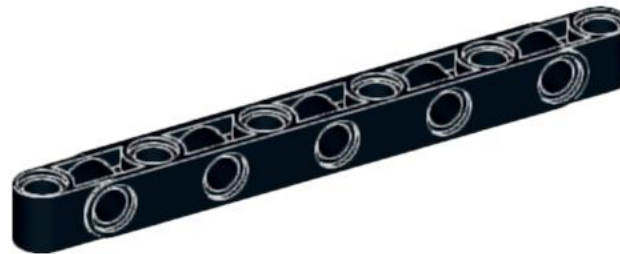
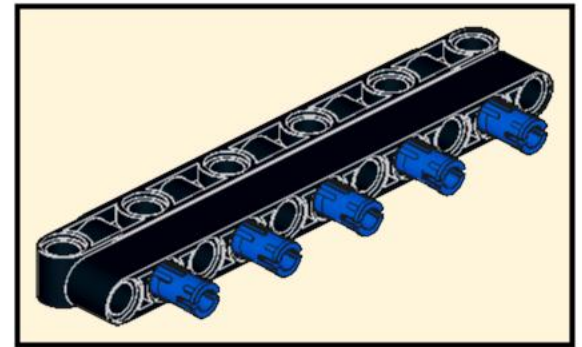
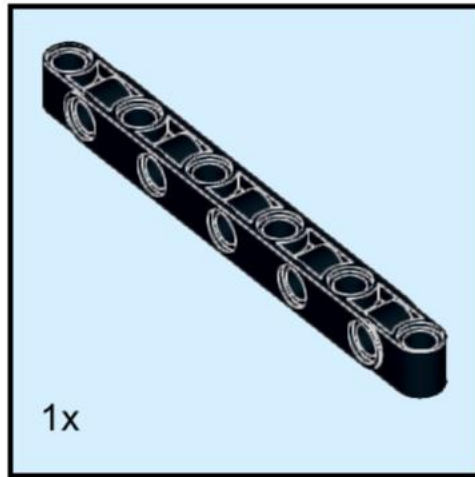


Hopefully it is obvious that these week assemblies will have be taken apart to install in the next step.

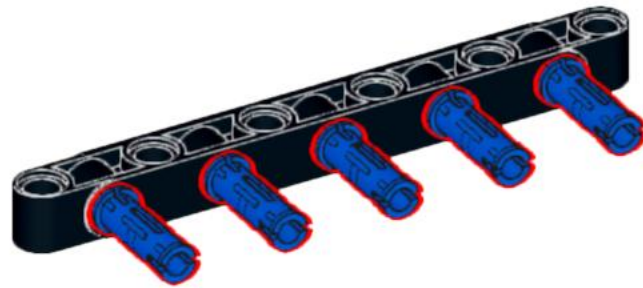
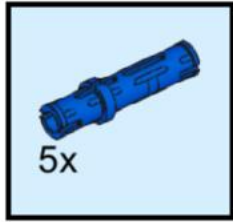
51



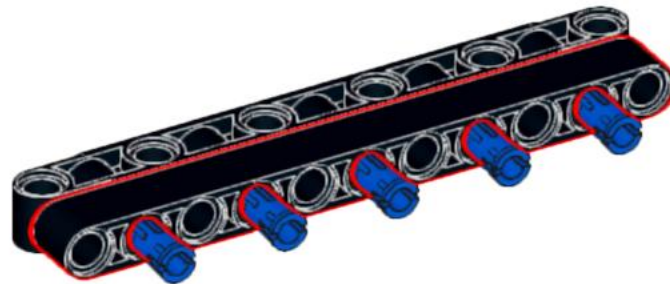
52



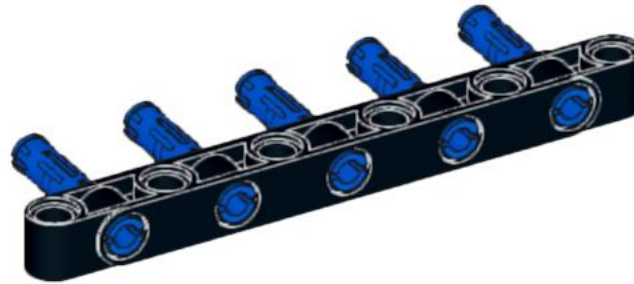
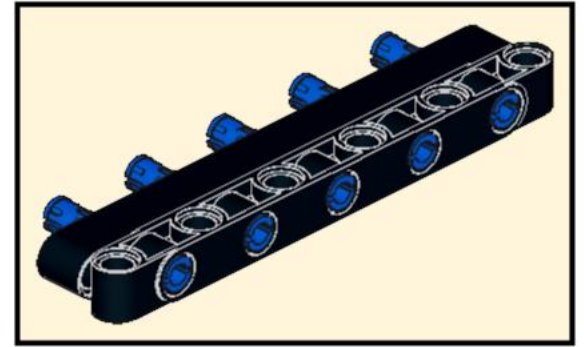
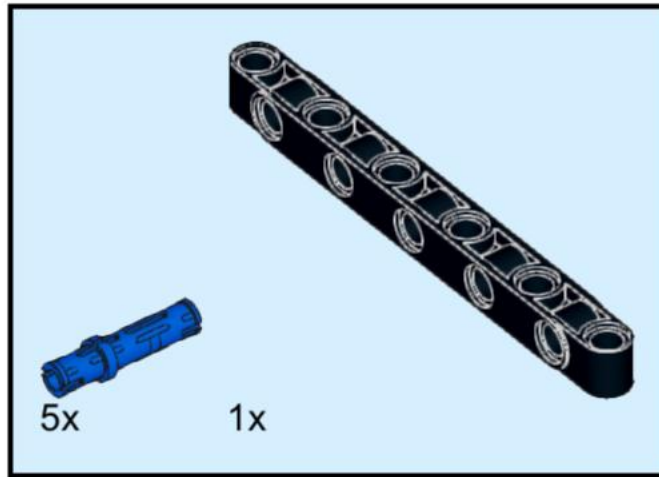
53



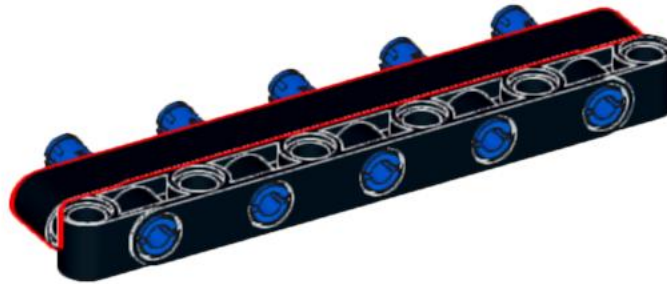
54



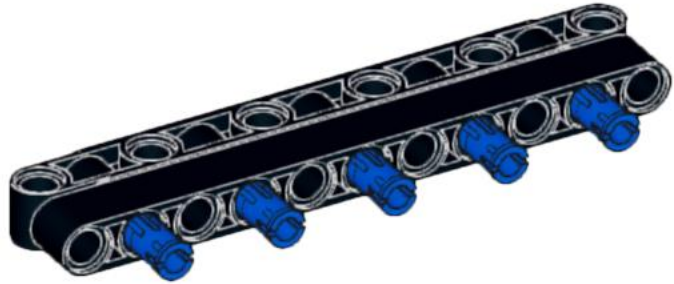
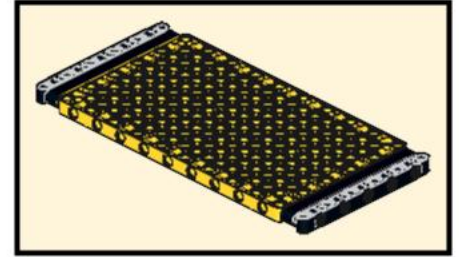
55



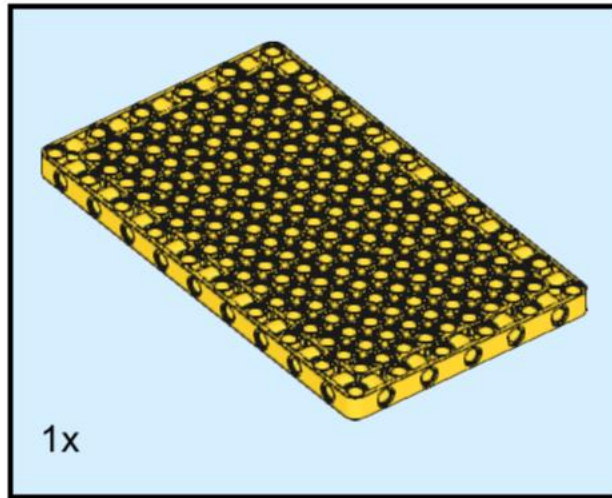
56



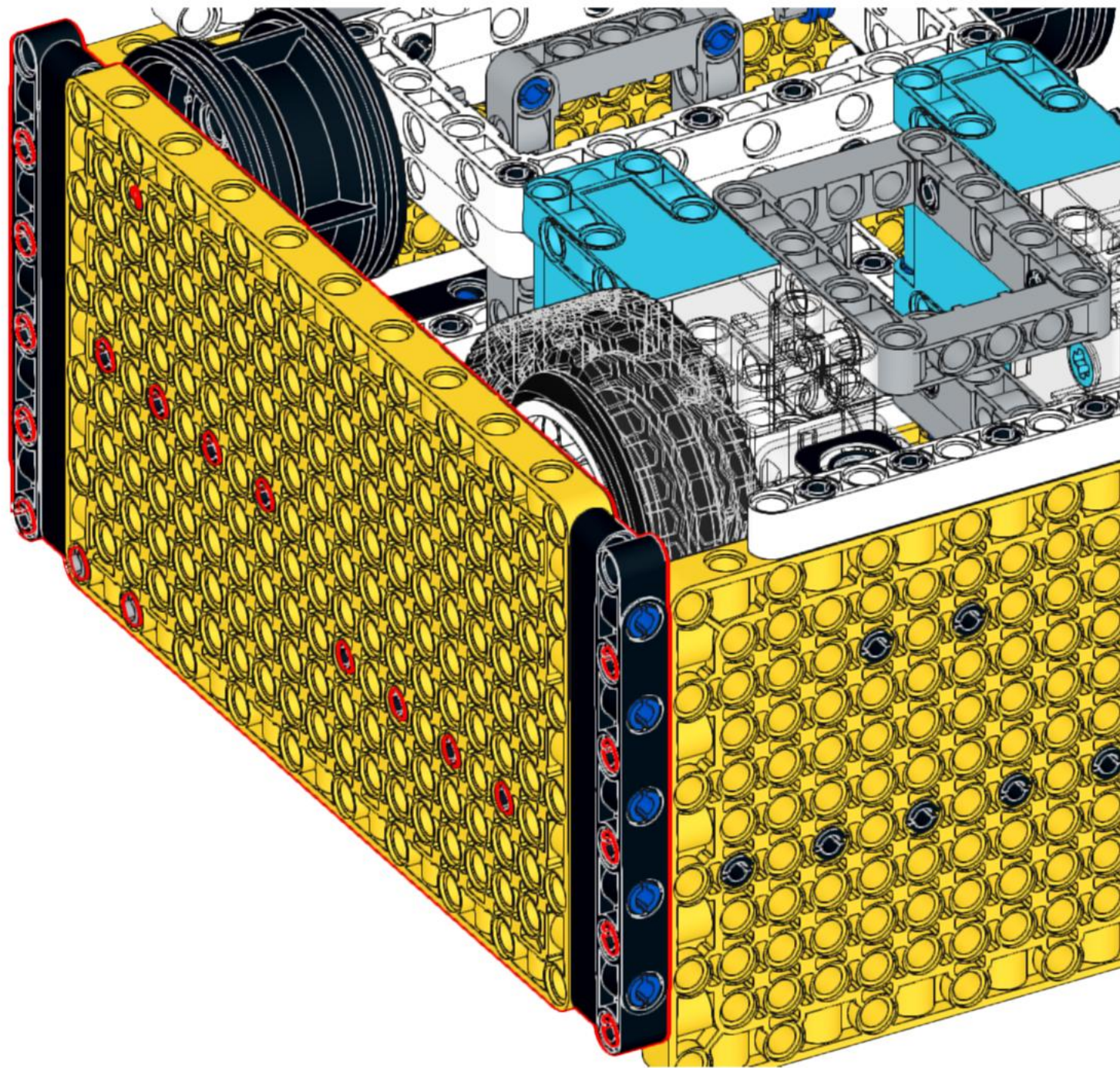
57



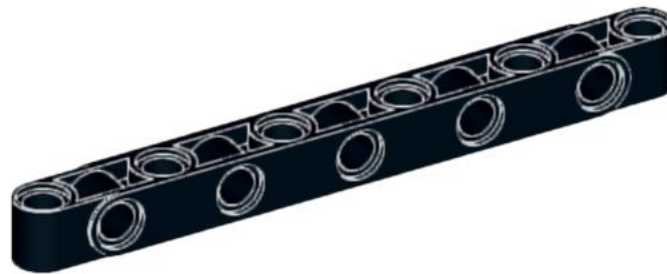
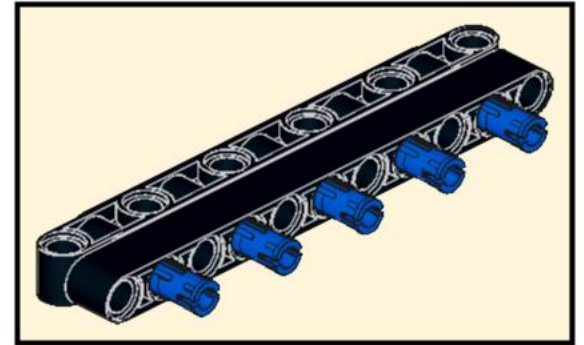
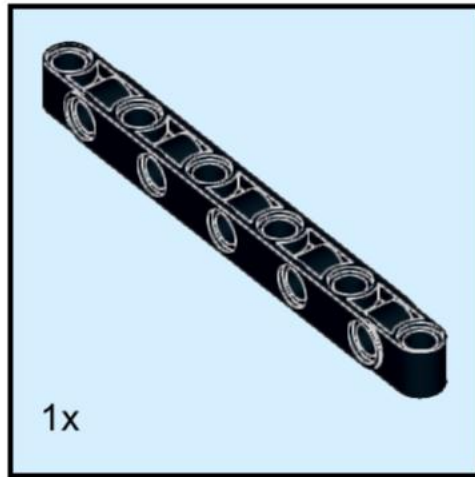
58



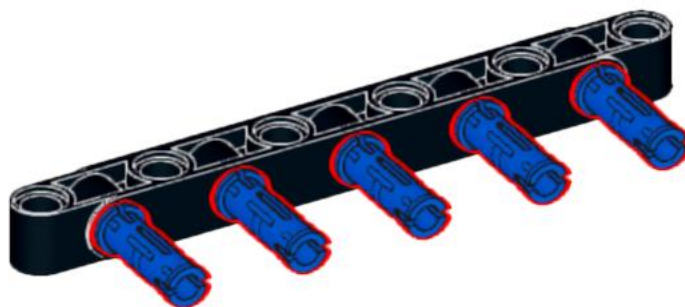
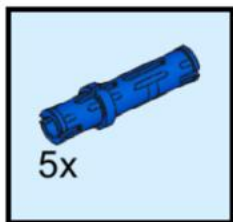
59



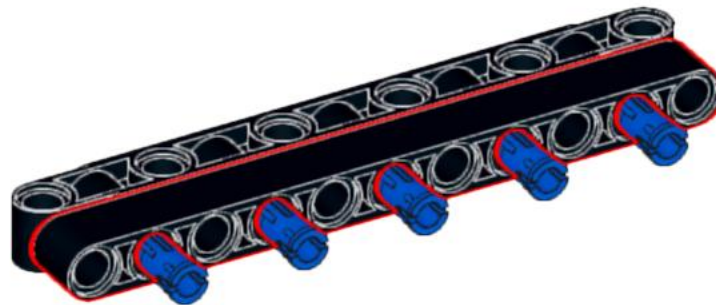
60



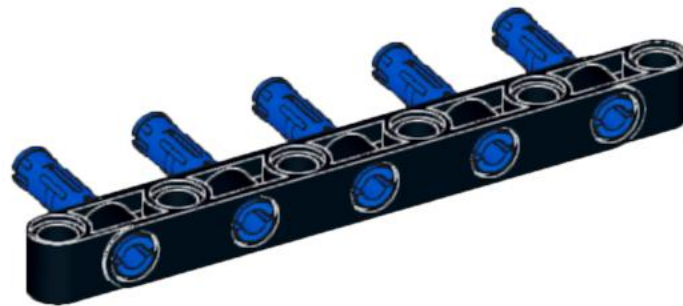
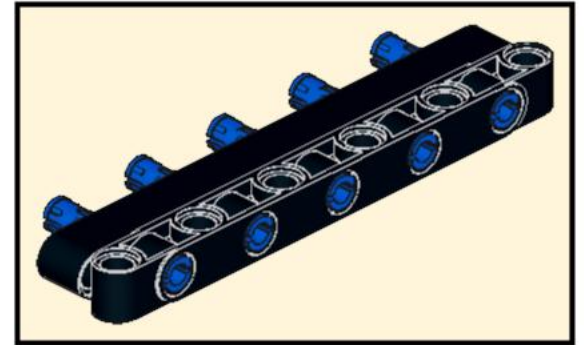
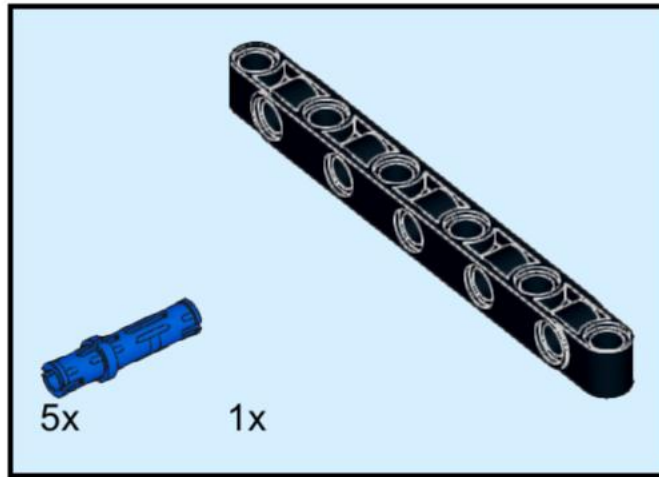
61



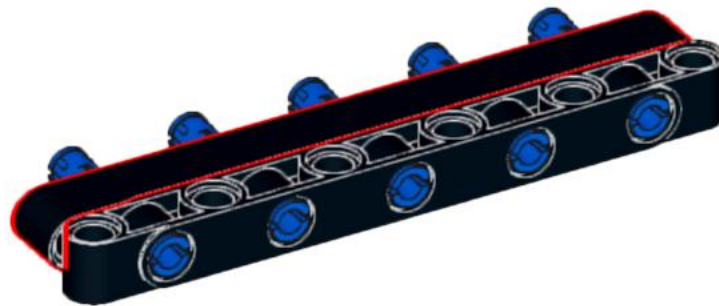
62



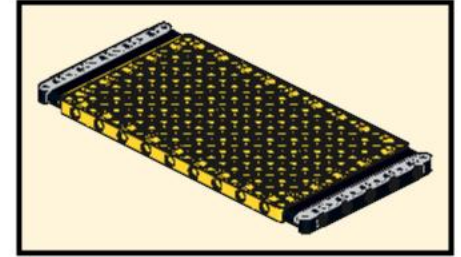
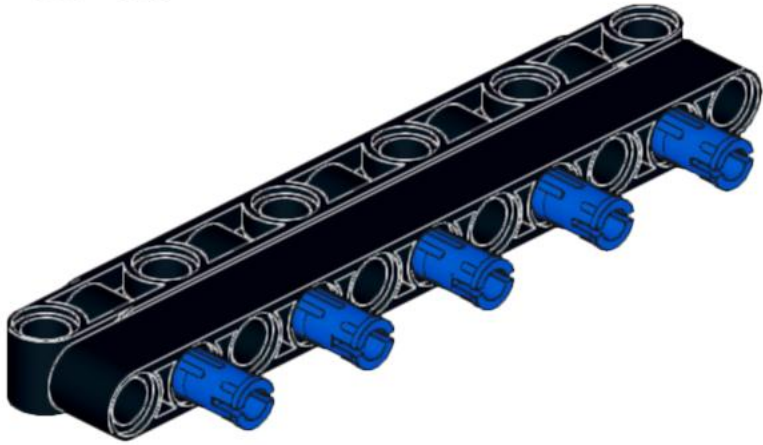
63



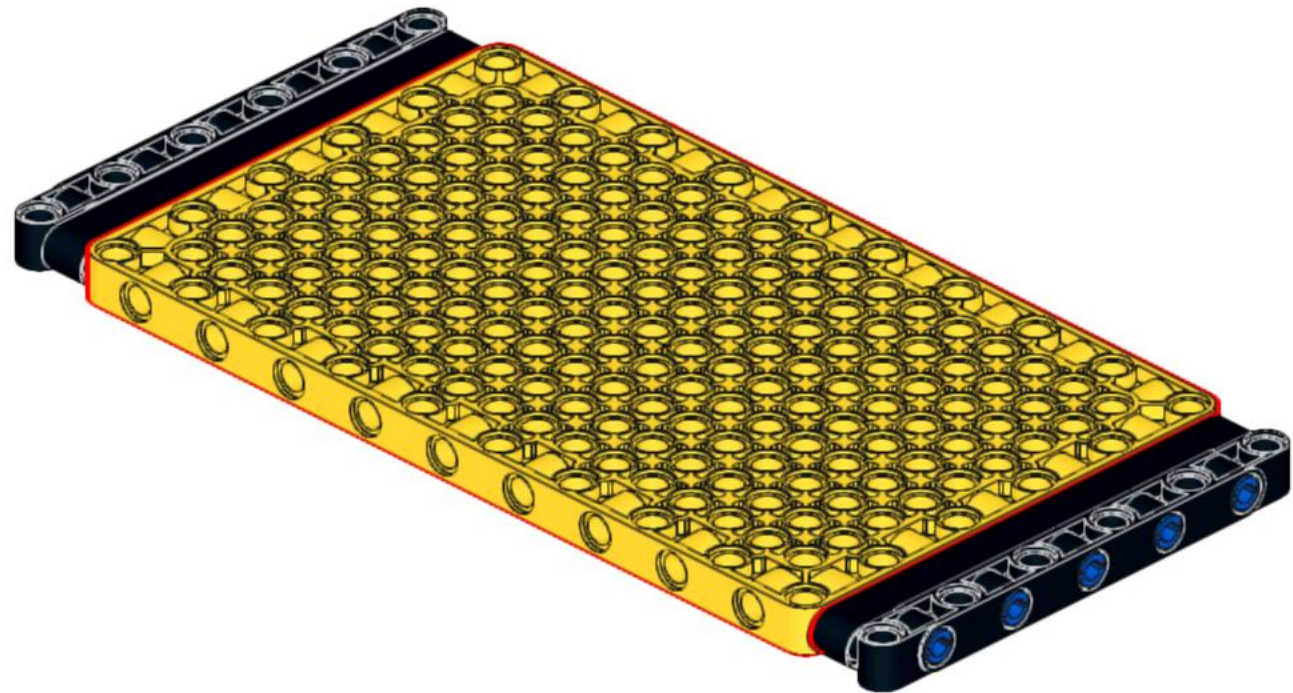
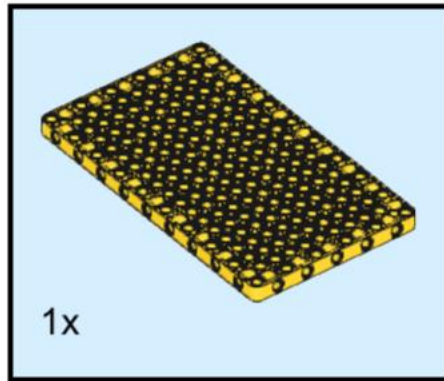
64



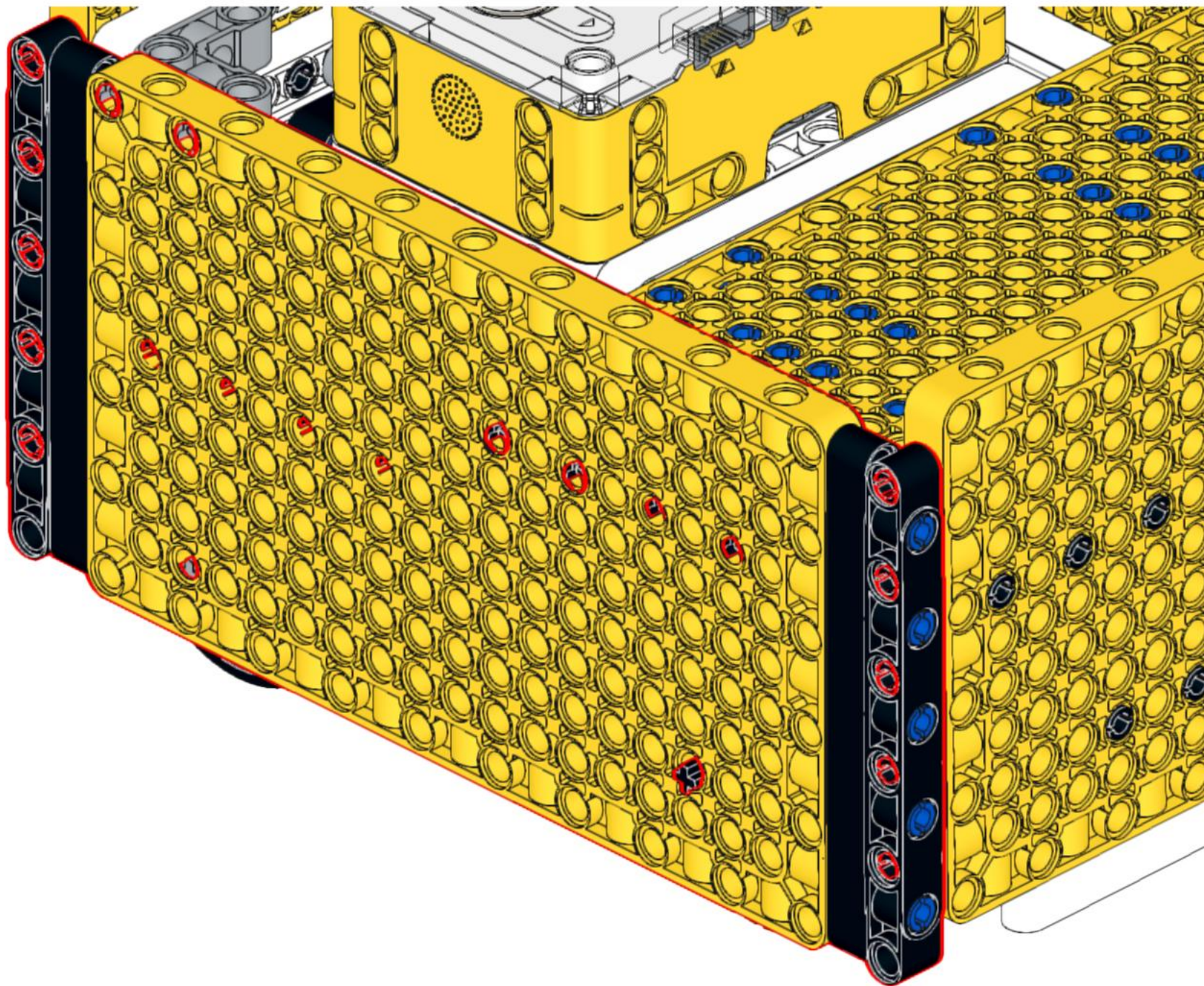
65



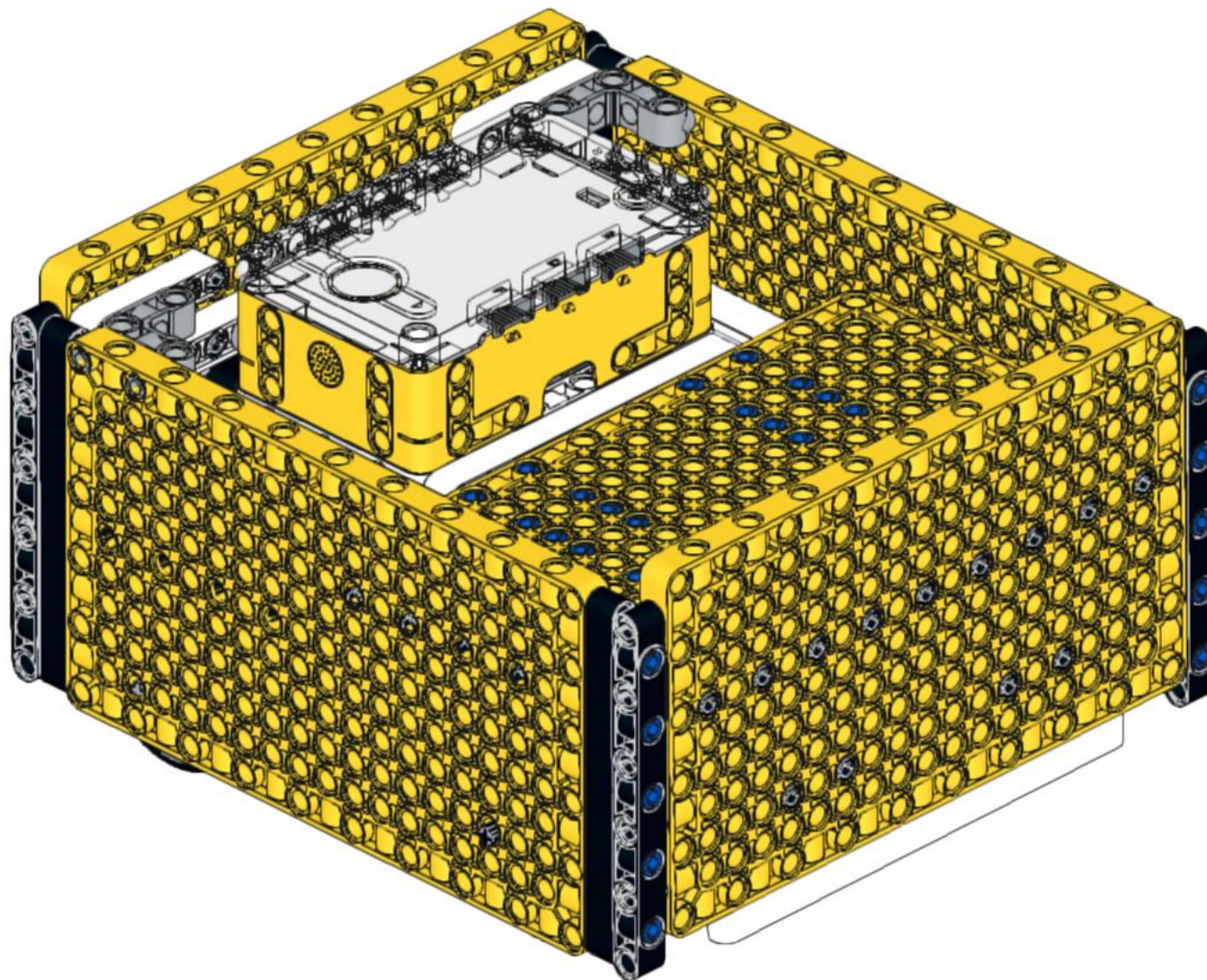
66



67



68



The core box robot is completed.

The next step is to install your motors. We recommend also using the large Spike motors due to substantially better torque over the medium Spike motors.

There are a substantial number of ways you can install the motors based on how the team wants its attachments to work.